



PE CIVIL-STRUCTURAL EXAM 01

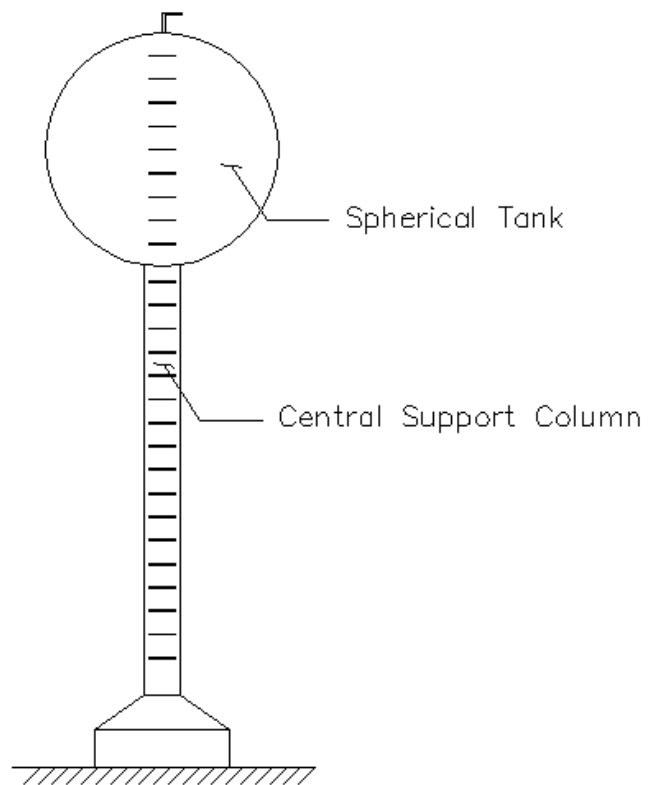
Table of Contents

Problems.....	2
Solutions.....	39

Problems

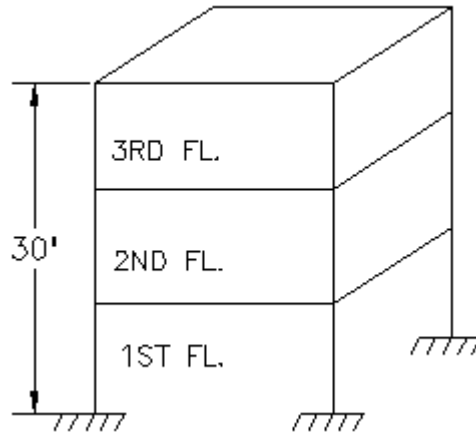
[Click here for Solutions](#)

1. An elevated reinforced concrete spherical water tank can store 2000 gallons of water. The walls of the tank are 6 inches thick. What is the total unfactored dead load acting on the supporting column?



- (A) 17 kips
- (B) 24 kips
- (C) 30 kips
- (D) 34 kips

2. A 3-story office building is to be constructed. The 1st floor will be used as an open parking lot and the 2nd and 3rd floors will be rented to two separate businesses. The businesses are not permanent, and the partition wall layouts will be changed to meet the requirements of the tenants. The typical floor height is 10ft and the typical floor area is 30ftX30ft. What is the total estimated partition wall load acting on the building?

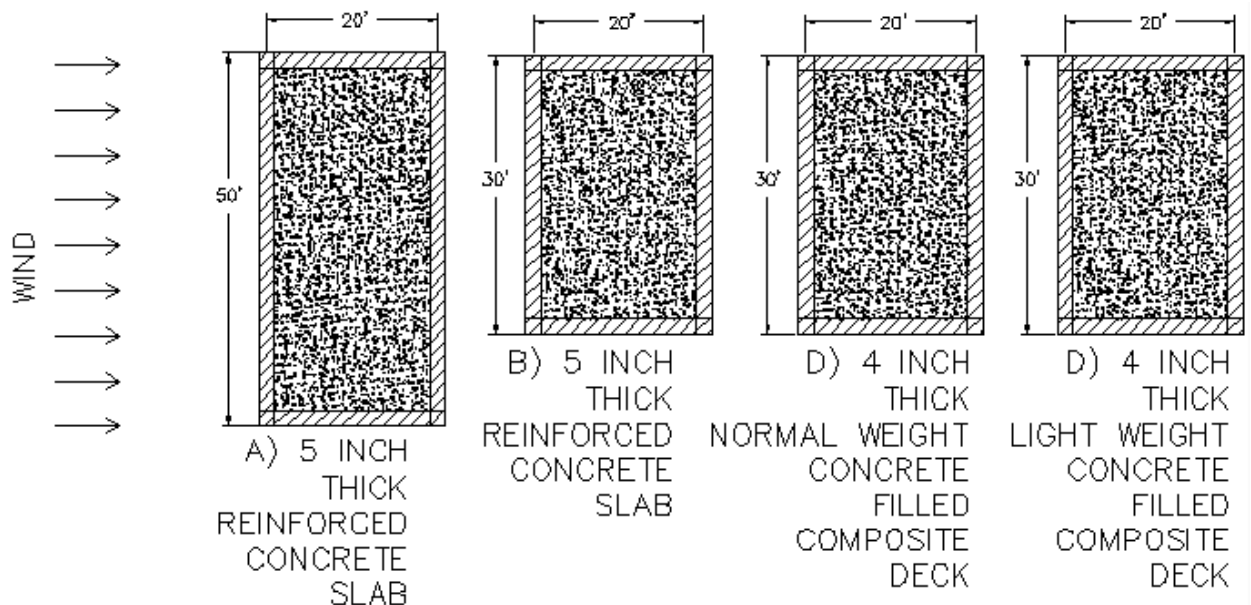


- (A) 13 kips
(B) 20 kips
(C) 27 kips
(D) 35 kips
3. A single story 50ftX100ft storage warehouse is built in Arizona. The warehouse will be heavily loaded with electronic goods. What is the total factored live load acting on the floor slab of the warehouse?
- (A) 390 kips
(B) 625 kips
(C) 1250 kips
(D) 2000 kips

4. For a deep foundation construction job, H piles are used in support of excavation systems. The section to be used is HP 10X57. The yield stress of the material is 50 ksi. What is the allowable driving load for the top driven pile?

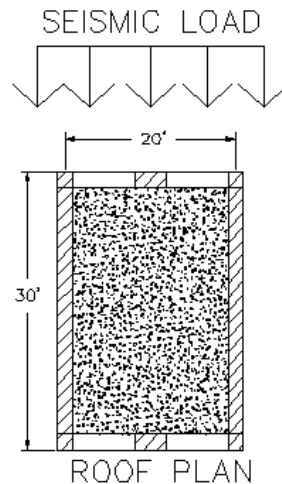


- (A) 500 kips
 (B) 750 kips
 (C) 1000 kips
 (D) 1500 kips
5. The plans below show Roof slabs for four different buildings. Which of the following cannot be idealized as a rigid diaphragm for analysis under wind loads per ASCE 7 -16?



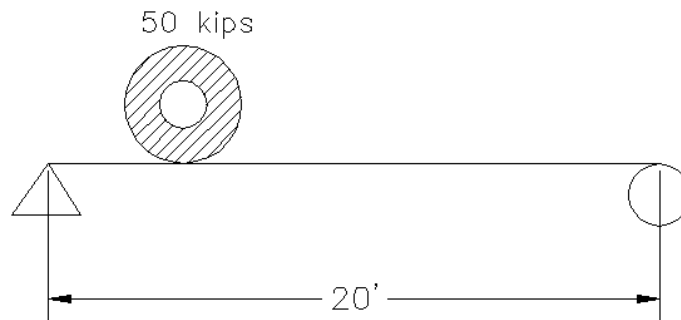
- (A) A
 (B) B
 (C) C
 (D) D

6. The plan below shows the roof plan of a one story building. The roof is a concrete slab, supported by masonry walls on longer sides and columns on shorter sides as shown by the hatches. During a seismic event, the lateral load acting at the roof slab level is 200 lb/ft and is along the direction as shown below. What is most nearly the loads shared by each wall?

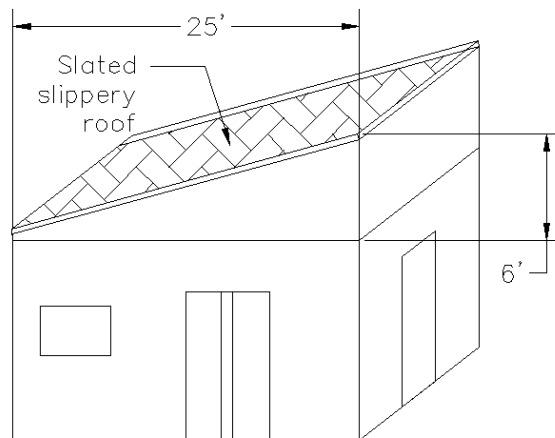


- (A) 1 kips
(B) 2 kips
(C) 3 kips
(D) 4 kips
7. A 3 story building is analyzed for seismic loads. The effective seismic weight of the structure is calculated as 800 kip. However, the design needs to be optimized. Which of the following can be removed from the calculation of seismic weight so that the seismic demands on the structural members can be reduced?
- (A) Weight of partition walls for the second floor office
(B) Weight of landscaping and permanent HVAC unit on the roof
(C) 20% of uniform design snow load
(D) 25% of floor live load for 'slab on grade' ground floor storage

8. A 50 kip wheel is moving along the span of a beam as shown below. What is the maximum bending moment acting in the beam due to the wheel load ?



- (A) 100 kip*ft
 (B) 250 kip*ft
 (C) 500 kip*ft
 (D) Cannot be determined
9. Calculate the design snow loads acting on the roof of a building located in Arlington, Washington shown below. The ground snow load is 17 psf. The building is classified as Risk Category II and the surface roughness category is C. The building is well heated and the roof is exposed on all sides.



- (A) 9.63 psf
 (B) 17 psf
 (C) 20 psf
 (D) 25psf

10. On a simply supported two lane bridge, the reaction due to design truck load is calculated as 40 kips. The design lane load reaction is calculated as 65 kips. What is the estimated braking force per AASHTO LRFD?

- (A) 2.5 kips
- (B) 5.25 kips
- (C) 10 kips
- (D) 26.25 kips

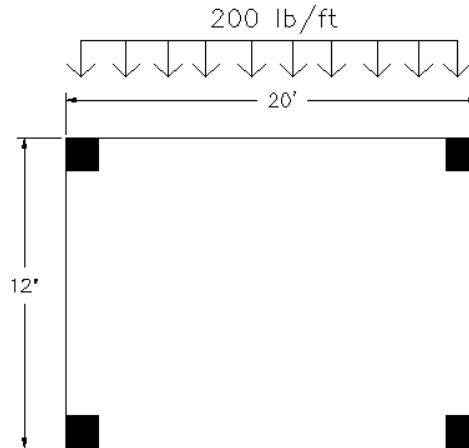
11. When designing support of excavation systems in a site with varying soil deposits, which of the following statements is correct to account for expansive pressure of soil?

- (A) Additional pressure due to swelling should be considered for all types of soil due to varying porosity,
- (B) Clayey soils are more susceptible to swelling due to varying moisture content during the course of excavation,
- (C) Sandy soils are more susceptible to swelling due to varying moisture content during the course of excavation,
- (D) Lateral soil pressure can be assumed at optimal moisture content if expansive soils are present at site.

12. A long line load of intensity 150 lb/ft acts on top of an embankment, parallel to a retaining wall. The line load acts at a distance 5ft from the face of the wall. The total depth of the wall is 20ft. What is the intensity of horizontal pressure on the wall acting 4ft below the top of the embankment due to this surcharge load per AASHTO LRFD specifications?

- (A) 10.13 lb/ft²
- (B) 11.38 lb/ft²
- (C) 12.88 lb/ft²
- (D) 14.69 lb/ft²

13. The plan shown below is a reinforced concrete floor and is supported by columns underneath. A lateral load of 200 lb/ft acts on the floor. What is the chord force in the diaphragm?

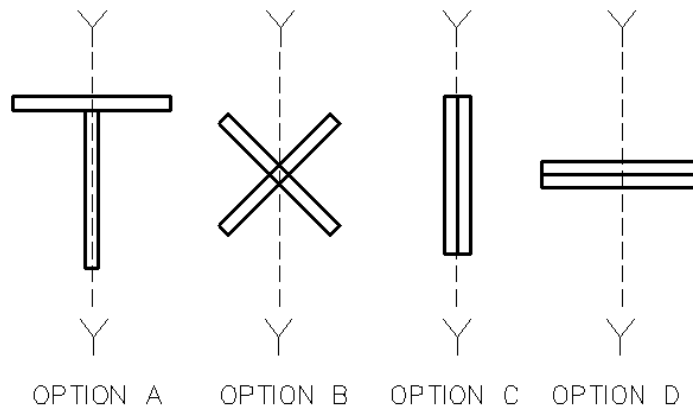


- (A) 200 lbs
- (B) 533.33 lbs
- (C) 833.33 lbs
- (D) 1000 lbs

14. Which of the following is not true about load combinations and associated limit states in AASHTO LRFD bridge design specifications?

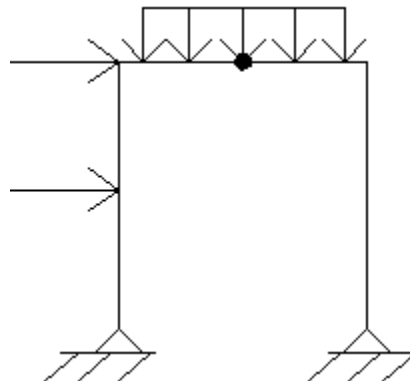
- (A) For less important bridges, the load factors recommended by chapter 3 can be scaled down.
- (B) For design using extreme event combinations, inelastic behavior which may restrict operation is not permitted
- (C) Typically, 'Strength IV' combination governs over 'Strength I' if the dead load effects are at least two times the live load effects,
- (D) The design forces for jacking in service shall not be less than 1.3 times the permanent load reaction at the bearing, adjacent to the point of jacking

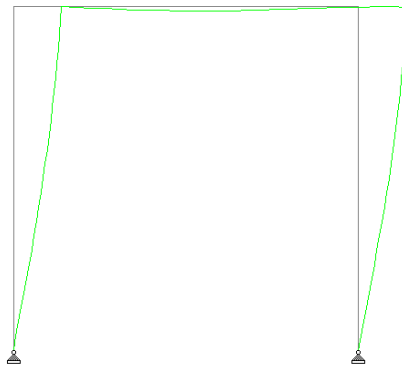
15. Two 1 inch by 12 in plates are used to form a cross section. The four shapes formed are shown as follows



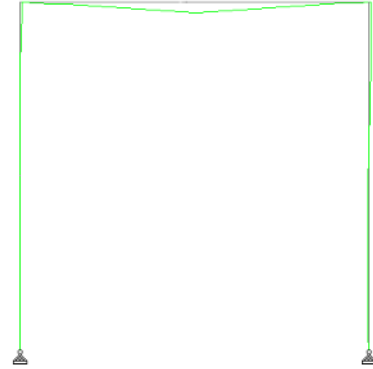
Which shape will have the highest flexural stiffness about Y-Y axis?

- (A) Option A
 - (B) Option B
 - (C) Option C
 - (D) Option D
16. For the given frame with a hinge at the center span of the beam, what is the true deflected shape?

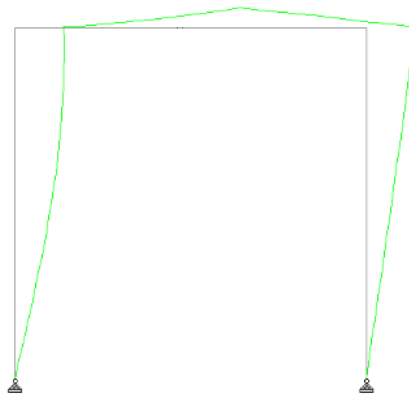




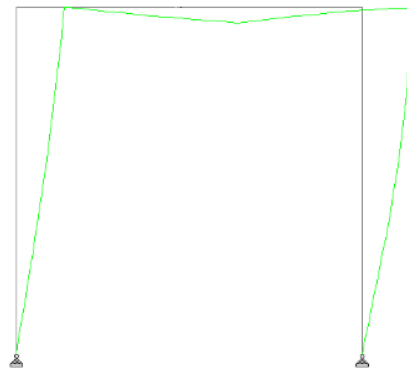
Option A



Option B



Option C



Option D

- (A) Option A
- (B) Option B
- (C) Option C
- (D) Option D

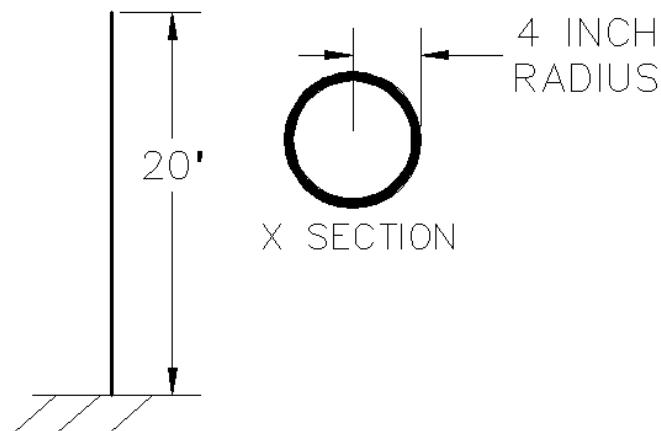
17. Which of the following is not true about influence line diagrams?

- (A) Influence line diagrams for reactions and internal forces for statically determinate structures are linear;
- (B) The shape of influence line diagram depends upon the system of units in the structure i.e. SI units or US customary units,
- (C) In case of a simply supported beam with overhangs, the influence lines for reactions and internal forces for section between supports should be constructed as for beam without overhangs and extended along the overhangs,
- (D) Mueller – Breslau's Principle can be used to obtain influence line diagrams for determinate and indeterminate structures.

18. Calculate the buckling load for a column that is 8 ft tall. Modulus of elasticity of the material is 29000 ksi . The cross section of the column is square with dimensions 4 inches by 4 inches. The column is free on one end and is pin supported (free to rotate) at the other end.

- (A) 200 kips
- (B) 300 kips
- (C) 400 kips
- (D) Cannot be determined

19. Calculate the critical buckling load for the column shown below. The column is 20 ft tall. Modulus of elasticity of the material is 29000 ksi . The cross section of the column is a circular tube with radius 4 inches and wall thickness 1 inch. The column is free on one end and fixed at the other end

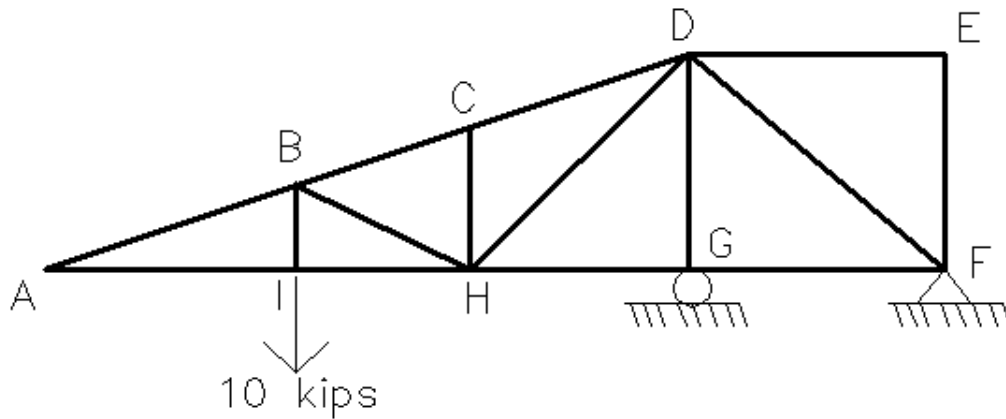


- (A) 150.8 kips
- (B) 170.7 kips
- (C) 300.2 kips
- (D) 682.9 kips

20. For axial load analysis, what is the primary assumption made in Euler's buckling theory to calculate failure load in slender columns?

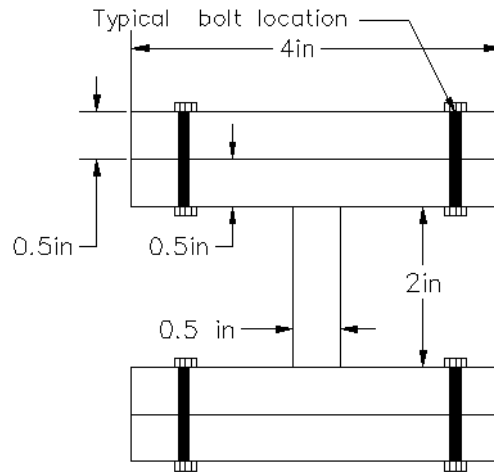
- (A) The self weight is considered for heavier columns,
- (B) The material of the column undergoes non linear behavior,
- (C) The column experiences considerable axial and shear deformations,
- (D) The column is subjected to purely axial load directly along the column axis.

21. Find the number of zero force members in the truss shown below. Horizontal load is acting on point D.

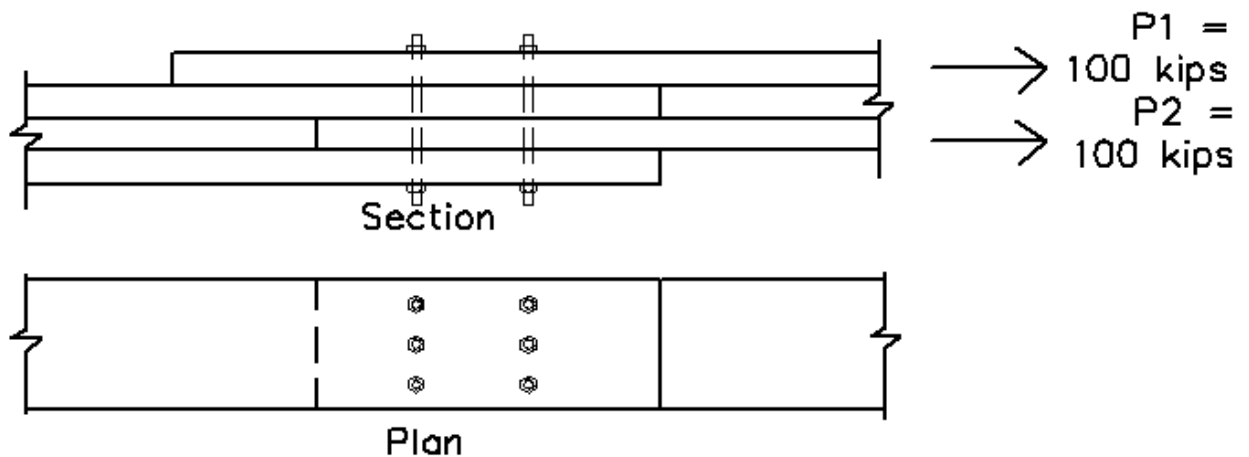


- (A) 3
- (B) 4
- (C) 5
- (D) 6

22. A built up girder system is shown below. The maximum shear force acting in the section is 15 kips. The top and bottom flange plates are connected by 2 bolts on either side of the web as shown below. What is the maximum shear load on each bolt if the bolts are spaced at 12 inches center to center along the span?



- (A) 12.2 kips
(B) 16.5 kips
(C) 21.6 kips
(D) 24.8 kips
23. The figure below shows the section and plan of a 4 plate connection system. The top plate is subjected to a load of 100 kips and the 3rd plate is also subjected to a load of 100 kips. There are a total of 6 bolts as seen in the plan. What is the maximum nominal shear stress acting on the bolts? The bolts are 1 inch in diameter.

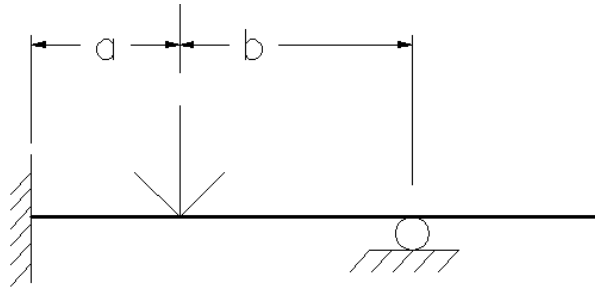


- (A) 21.23 ksi
(B) 42.46 ksi
(C) 63.69 ksi
(D) 84.92 ksi

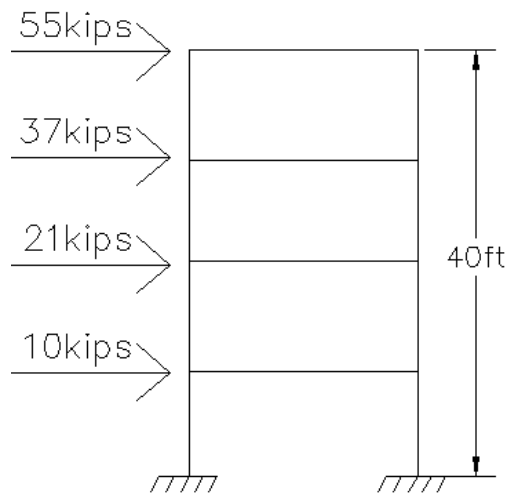
24. A propped cantilever beam with a concentrated load is shown in the figure below. To reduce the maximum shear at fixed support, what design intervention can be used?

- I) Reduce span b
- II) Reduce the magnitude of load
- III) Increase the depth of beam section

- (A) I, II and III
- (B) I and II
- (C) II only
- (D) II and III

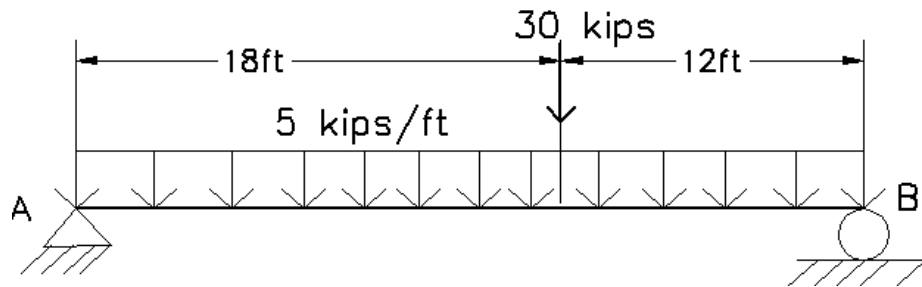


25. The figure below shows lateral seismic load distribution on each floor level. What is the total overturning moment at the base of the building? Assume all story heights are equal.



- (A) 2365 kip*ft
- (B) 2870 kip*ft
- (C) 3292 kip*ft
- (D) 3830 kip*ft

26. For the beam shown below, what is the location of maximum bending moment from support A?



- (A) 17.4 ft
- (B) 18 ft
- (C) 18.6 ft
- (D) 19.2 ft

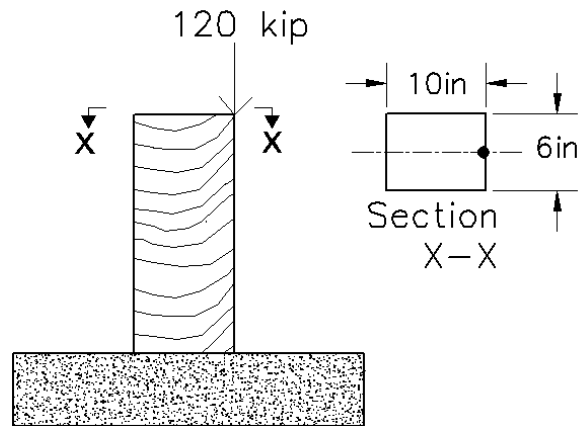
27. The state of stresses in the wall of a vessel is given as follows:

- a. Compressive stress (along XX) = 10 ksi
- b. Tensile stress (along YY) = 25 ksi
- c. Shear stress = 15 ksi

Calculate the principal stresses acting on the body.

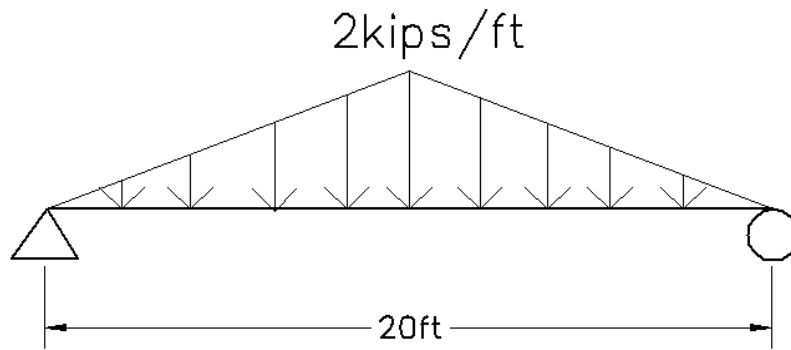
- (A) 30.5 ksi compressive, 15.5 ksi tensile
- (B) 15.5 ksi compressive, 30.5 ksi tensile
- (C) 42.2 ksi compressive, 13.8 ksi tensile
- (D) 13.8 ksi compressive, 42.2 ksi tensile

28. A wooden post rests on a concrete base. A 120 kip load acts on the top of the post at the center of the right edge of the post, as shown in section X-X. What is the maximum compressive stress exerted by the post on the concrete base?



- (A) 2 ksi
(B) 4 ksi
(C) 8 ksi
(D) 10 ksi
29. A wooden block is fixed rigidly at the base. A tangential load is applied to the upper surface of the block. The deflection caused at the top of the block is due to
- (A) Shear and Bending moment,
(B) Bending moment only,
(C) Shear and torsional moment,
(D) Shear only,

30. A simply supported beam is subjected to a uniformly varying load as shown below. The moment of inertia of the beam section is 60 in^4 and the modulus of elasticity is 29000 ksi . What is the intensity of an equivalent uniformly distributed load that causes the same deflection in the beam?

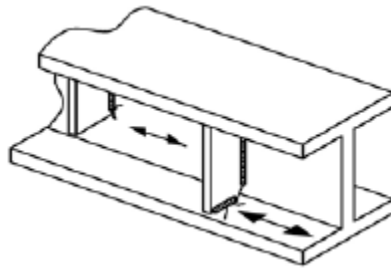


- (A) 0.7 kip/ft
(B) 1 kip/ft
(C) 1.3 kip/ft
(D) 1.5 kip/ft
31. A simply supported beam is subjected to uniformly distributed load. The beam has a rectangular cross section. If the intensity of loading is halved and the depth of beam is doubled, what is the ratio of maximum deflections of the new beam and the original beam?
- (A) One third
(B) One quarter
(C) One eighth
(D) One sixteenth
32. Two beams with identical cross sections and lengths are manufactured with two materials. One of them is manufactured with gray cast iron and another is manufactured with mild steel. Both of them are fixed in two ends identically and identical uniform loading is applied. Which of the following is true about their deflections?
- (A) Gray cast iron beam will deflect less than mild steel beam,
(B) Mild steel beam will deflect less than gray cast iron beam,
(C) For identical cross section, loading and statically determinate conditions, the deflections will be equal for both materials.
(D) Deflections will depend upon the allowable yield stresses of both materials.

33. Which of the following structural members are most susceptible to lateral torsional buckling?

- (A) Unbraced beams subjected to significant axial loads,
- (B) Unbraced columns subjected to significant axial loads,
- (C) Unbraced beams subjected to significant bending moments,
- (D) Roof truss members subjected to significant wind loading.

34. A gusset plate is welded to the web of a bridge girder as shown below, near an interior support. The span of the girder is 60 ft and is a part of a continuous girder system. Calculate the minimum fatigue resistance of the welded plate. The single lane average daily truck traffic is 2000 vehicles per day.



- (A) 3.77 ksi
- (B) 4.21 ksi
- (C) 5.38 ksi
- (D) 7.63 ksi

35. A real estate agency hires a class B contractor to construct a 12 story building. Which of the following parties can qualify as a special inspector?

- (A) Department of building official,
- (B) Engineer of record of the building to be constructed,
- (C) Registered professional employed by the contractor to supervise construction,
- (D) An employee from the subcontractor's team who has necessary experience and complete information on material specifications to be used in the project.

36. What is the contractor's role during the submittal process?

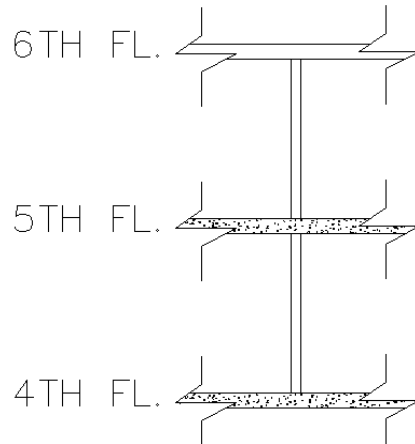
- (A) To create and approve all submittal documents,
- (B) To compile and review all submittal documents before forwarding it to the building engineer of record,
- (C) To revise and document design assumptions based on discrepancies on site and capabilities of subcontractor's shop,
- (D) To deliver Department of Building approved design drawings to the engineer of record.

37. Guy ropes are used to stabilize a bridge falsework system. The maximum pretension in the guy rope is 2.5 kips. The guy rope is inclined at 45 degrees to the horizontal and is anchored to a 4 ksi concrete slab. The table below shows ultimate forces that the anchors can withstand in lbs. A factor of safety of 2 can be used to convert the ultimate forces to allowable forces. What is the minimum diameter of the anchor that can be used?

Anchor Dia. (in)	Required Torque (ft-lb)	Embedment depth (in)	$F'_c = 2000$ psi		$F'_c = 4000$ psi	
			Tension	Shear	Tension	Shear
$\frac{1}{4}$	8	$2\frac{1}{8}$	2,260	1,680	3,300	1,680
$\frac{3}{8}$	25	4	4,800	4,000	5,940	4,140
$\frac{1}{2}$	55	6	5,340	7,240	9,640	7,240
$\frac{5}{8}$	90	$7\frac{1}{2}$	7,060	9,600	15,020	11,900

- (A) 1/4 in
- (B) 3/8 in
- (C) 1/2 in
- (D) 5/8 in

- 38.** The 6th floor of a concrete building is about to be cast. The shoring posts on each floor are laid at a spacing 5 ft X 5 ft. The thickness of all floor slabs are 6 inches. The weight of formwork and falsework can be assumed to be 15 psf. Density of concrete is 150 pcf. The construction load on the 5th floor is 50 psf and on the 6th floor is 75 psf. Assuming that all loads from the 6th floor and 5th floor are carried by formwork and shoring posts, what will be the maximum load acting on the shoring posts under 5th floor?



- (A) 4.7 kips
(B) 5.8 kips
(C) 6.3 kips
(D) 7.6 kips
- 39.** For a concrete wall formwork, the plywood sheathing is supported by vertical posts and horizontal wales. The wales are spaced vertically at 24 inches center to center and the vertical posts are spaced 30 inches center to center horizontally. A bundle of six tie wire strands are used to tie the wales and posts together. Every alternate vertical post is tied to the wales. If the rate of concrete placement is 3 inches per hour and the lateral pressure exerted by the concrete on the forms is 400 psf, what is the maximum load acting on the tie wire?
- (A) 2 kips
(B) 4 kips
(C) 6 kips
(D) 8 kips

40. A foreman at a demolition site is exposed to the following noise levels during an 8 hour working shift:

8 a.m. to 12 p.m. = 90 dBA

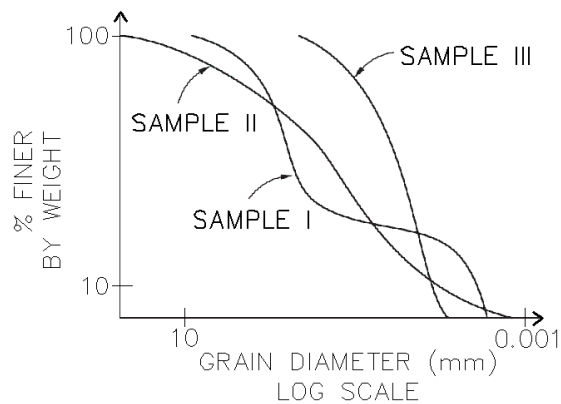
12 p.m. to 2 p.m. = 100 dBA

2 p.m. to 4 p.m. = 95 dBA

What is the noise dose per OSHA regulations?

- (A) 80%
- (B) 90%
- (C) 100%
- (D) 200%

41. The graph below shows particle size distribution for three soil samples I, II and III. Which of the following denotes the correct grading of the samples?



- | | | |
|---------------------|-----------------|------------------|
| (A) I - Well graded | II- Gap graded | III- Uniform |
| (B) I - Gap graded | II- Well graded | III- Uniform |
| (C) I - Uniform | II- Well graded | III- Gap graded |
| (D) I - Gap graded | II- Uniform | III- Well Graded |

42. Standard penetration test was conducted at a site. At the required depth , the effective overburden pressure was 4 ksf and the energy corrected N_{60} value was 20 blows. What is the estimated in situ dry unit weight of the soil sample?

Data obtained from lab tests are:

$e_{max} = 0.4$ $e_{min} = 0.2$ specific gravity of soil sample , $G_s = 2.7$

- (A) 115.2 pcf
- (B) 120.8 pcf
- (C) 125.4 pcf
- (D) 131.6 pcf

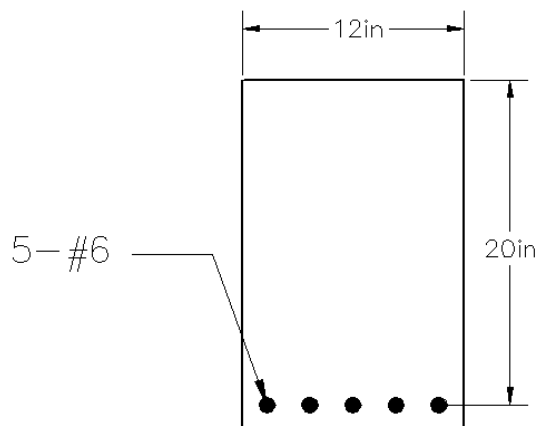
43. The test results for standard 4 inX 8 in cylinders from a job site for the first six days of pour, cured for 28 days yielded the following results:

Day number	Compressive strength
1	3.12 ksi
2	3.4 ksi
3	4 ksi
4	4.1 ksi
5	2.98 ksi
6	3.9 ksi

If the design strength of concrete mix is 3500 psi, which of the following is true?

- (A) Mix is not compliant because tests from Day 1, Day 2 and Day 5 is below 3500 psi,
 - (B) Mix passes because overall average of the tests is above 3500 psi,
 - (C) Mix passes because the average of any three consecutive tests is above 3500 psi,
 - (D) Mix is not compliant because the test result from day 5 is below 3000 psi.
44. For the section shown below, what is the strength reduction factor for flexure per ACI 318-14? Grade of steel is 60 ksi and concrete strength is 4000 psi.

- (A) 0.65
- (B) 0.75
- (C) 0.85
- (D) 0.9



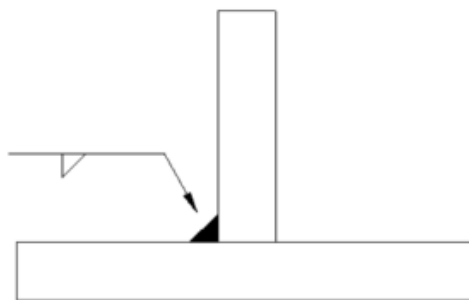
45. What is the cracked moment of inertia of a singly reinforced rectangular section with depth 24 in and width 12 in?. The beam is reinforced with 4#6 bars at the bottom and the effective depth is 22 in. The modulus of elasticity of concrete is 3600 ksi and the modulus of elasticity of rebar is 29000 ksi.

- (A) 924 in⁴
- (B) 1875 in⁴
- (C) 4494 in⁴
- (D) 4804 in⁴

46. An ASTM A307 bolt is subjected to a shear stress of 10 ksi. What is the reduced allowable tensile stress in the bolt material when subjected to shear tension interaction?

- (A) 8 ksi
- (B) 12 ksi
- (C) 25 ksi
- (D) 35 ksi

47. The strength of filler metal for the weld below is 70 ksi. The weld size is 3/8 inch. What is the allowable strength of the weld per unit length when it is loaded transversely to its longitudinal axis?



- (A) 5.57 kip/in
- (B) 8.35 kip/in
- (C) 11.14 kip/in
- (D) 16.7 kip/in

48. Which of the following is not true about the properties of cold formed steel ?

- (A) Cold formed steel exhibits anisotropic material behavior due to the forming process,
- (B) Cold formed steel sections develop longitudinal residual stresses due to differential plastic deformation during forming,
- (C) Cold forming process induces plastic deformation, particularly at the corners, leading to strain hardening which leads to increased yield strength in these regions,
- (D) Because of the difficulty of cold working instead of being hot rolled, the dimensional tolerances for cold formed steel are higher than that for hot rolled sections.

49. A simply supported wooden floor beam is subjected to a uniformly distributed live load of 50plf. The center to center distance between the end bearing plates is 12 ft. What is the minimum required moment of inertia of a Select structural Douglas Fir Larch dimensional lumber to satisfy IBC deflection limits? Assume $C_i = 1$ and dry conditions, service temperature less than 100°F.

- (A) 31 in⁴
- (B) 35 in⁴
- (C) 41 in⁴
- (D) 50 in⁴

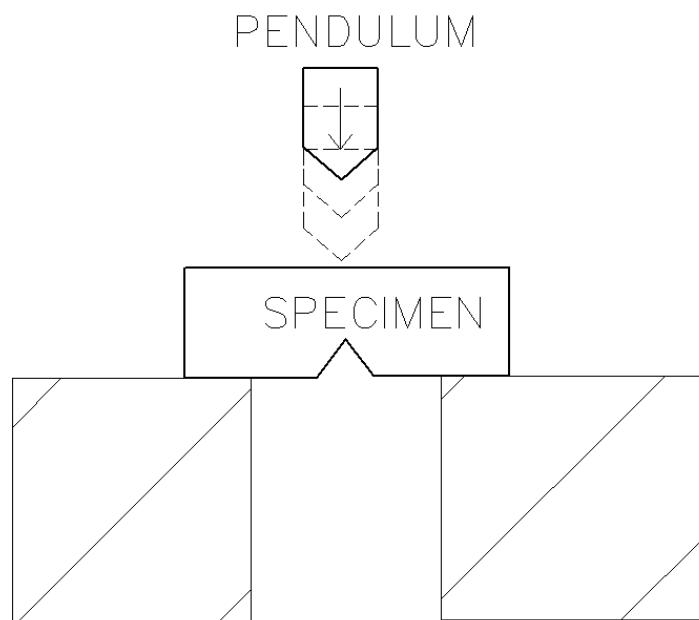
50. The specified compressive strength of a concrete masonry unit is 1500 psi. The masonry unit is constructed with Type M mortar using ordinary portland cement. The unit is unreinforced, fully grouted. What is the allowable flexural tensile stress normal to the bond ?

- (A) 33 psi
- (B) 40 psi
- (C) 53 psi
- (D) 65 psi

51. What is the fire resistance rating of a concrete masonry wall constructed of calcareous gravel that has an equivalent thickness of 5 inches?

- (A) 1.5 hour
- (B) 2.25 hour
- (C) 2.75 hour
- (D) 3 hour

52. A V- notched steel specimen is centered on simple supports and is struck with a falling pendulum. The kinetic energy of the falling pendulum to cause failure of the specimen is recorded at standard temperature. Which of the following tests is represented by this procedure?



- (A) Toughness test
- (B) Hardness test
- (C) Fatigue test
- (D) Fracture test

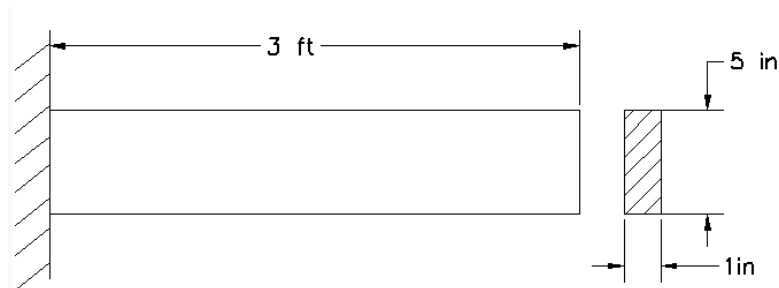
53. A simply supported steel beam has a span of 25 ft and is used as a floor member. The beam is subjected to a concentrated service live load of 80 kips at the midspan and bends about the major axis. What is the lightest A992 section that will satisfy the IBC 2018 serviceability requirements?

$E = 29000 \text{ ksi}$



- (A) W18X86
- (B) W18X97
- (C) W18X106
- (D) W18X119

54. A cantilevered horizontal bar of cross section 5 in X 1 in is cantilevered with $L = 3 \text{ ft}$ as shown below. The yield strength of the material is 50 ksi and $E = 29000 \text{ ksi}$. What is the allowable flexural strength of the bar? Assume $C_b = 1$.



- (A) 12.1 kip*ft
- (B) 13.2 kip*ft
- (C) 14.9 kip*ft
- (D) 15.5 kip*ft

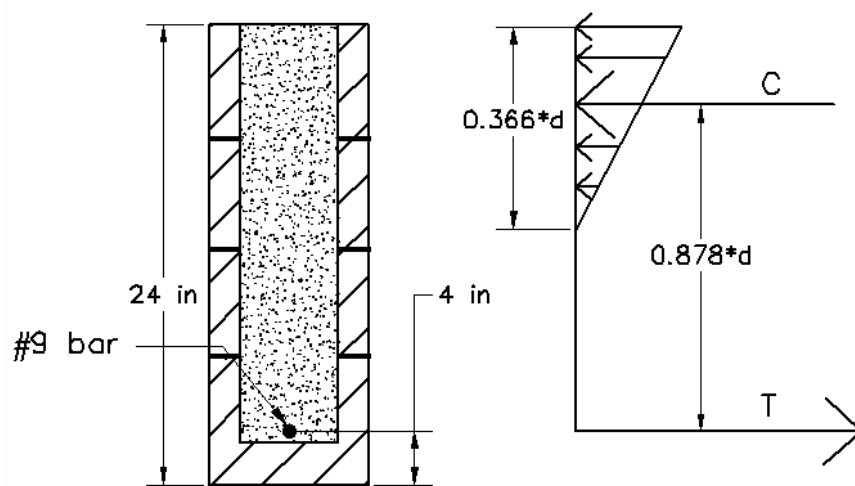
55. What is the allowable bending moment for the reinforced masonry beam section below? The section is flexurally reinforced with one no. 9 bar .

Section properties:

Rebar $f_y = 60$ ksi

Masonry $f'_m = 2000$ psi

Width of the section = 7.625 in.



- (A) 20 kip*ft
- (B) 27.5 kip*ft
- (C) 32.25 kip*ft
- (D) 36.75 kip*ft

56. Which of the following masonry beams can be considered a deep beam?

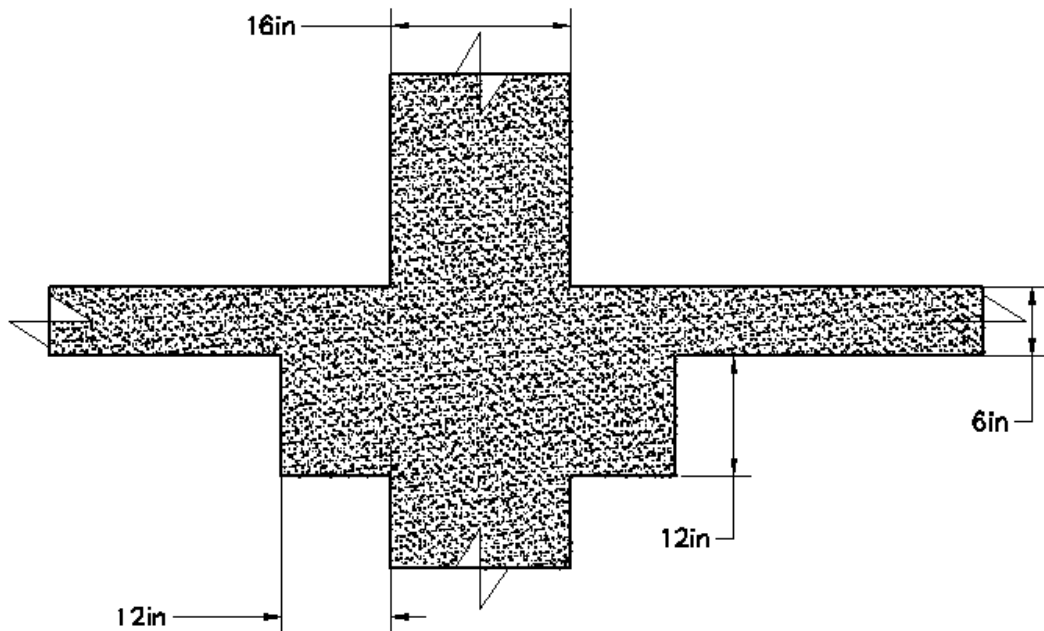
- I) A simply supported masonry beam with effective span 10 ft and depth 65 in
- II) A continuous span masonry beam with span length 15 ft and depth 65 in
- III) A continuous span masonry beam with section width 7.625 in and depth 72 in

- (A) I only
- (B) I and II
- (C) III only
- (D) I, II and III

57. An Eastern white pine select structural grade, simply supported rafter spans 15 ft. The rafter is spaced 22 inches center to center. The rafter is subjected to a uniform snow load of 50 plf and is loaded along the thickness. What is the minimum required section modulus of the lumber? Assume dry condition and normal temperature.
 $C_L = C_i = C_f = 1$

(A) 10.2 in^3
 (B) 12.3 in^3
 (C) 14.2 in^3
 (D) 15.7 in^3

58. A two way flat slab is strengthened around an interior column using drop panels as shown below. Both the column and the panel are square in dimension. If the nominal two way shear strength of the concrete is 250 psi, what is the design two way shear strength of the slab at the column location? Assume effective cover of tension rebars in the slab is 1.25 in

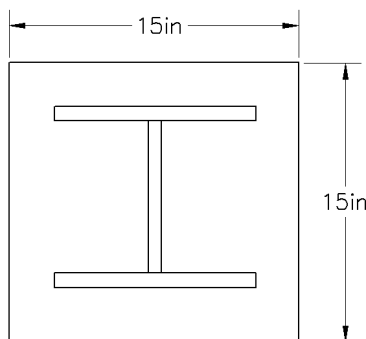


(A) 98 kips
 (B) 115 kips
 (C) 160 kips
 (D) 189 kips

59. A fixed-fixed HSS 8X8X1/2 column is subjected to concentric axial compression. The height of the column is 19.25 ft. What is the maximum allowable axial compression that can be resisted by the column? $F_y = 70$ ksi and $E = 29000$ ksi



- (A) 200 kips
(B) 300 kips
(C) 440 kips
(D) 710 kips
60. A W section is centered on a 15 in X 15 in base plate as shown below. The overall depth of the section, $d = 11$ in and flange width, $bf = 12$ in. What is the minimum required thickness of the base plate to support a factored axial load of 650 kips? $F_y = 50$ ksi

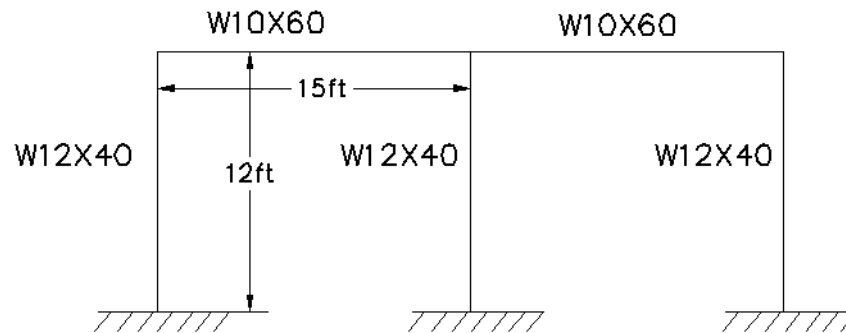


- (A) 1 in
(B) 1.125 in
(C) 1.375 in
(D) 1.625 in

61. For the moment resisting frame below, what is the effective length factor for the leftmost column?

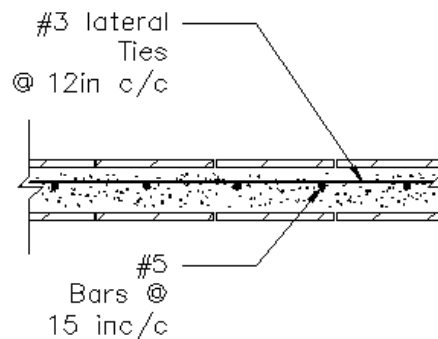
$E = 29000 \text{ ksi}$

$F_y = 50 \text{ ksi}$



- (A) 1
(B) 1.2
(C) 1.3
(D) 2
62. A masonry wall section is shown below. The thickness of the section is 7.625in and the slenderness ratio $H/r = 100$, The vertical reinforcement is Gr. 60 #5 bars spaced at 15 in c/c. If the compressive strength of the masonry is 2000 psi, what is the allowable axial load in the wall section?

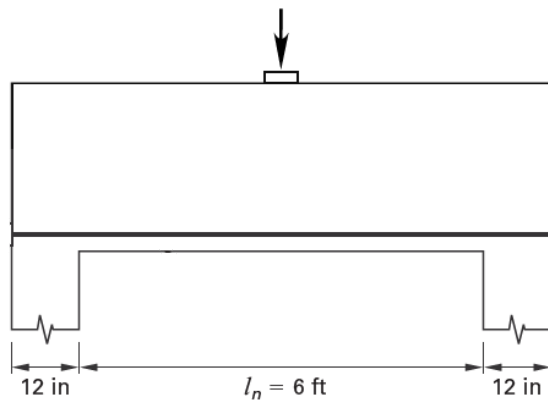
- (A) 22 kips/ft
(B) 38 kips/ft
(C) 45 kips/ft
(D) 65 kips/ft



63. What is the maximum number of #9 longitudinal bars that can be used for a 10 inch X 10 inch RCC column section per ACI 318-14?

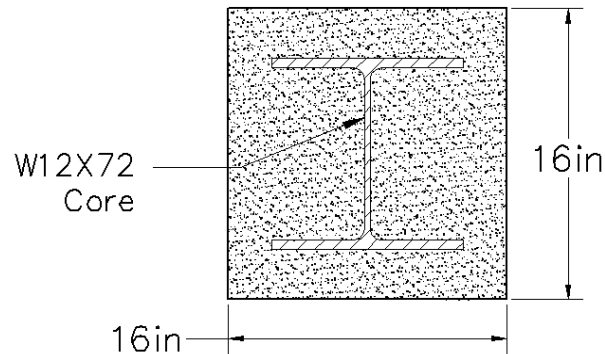
- (A) 4
- (B) 6
- (C) 8
- (D) 12

64. A deep beam has a clear span of 6 ft and bears on two 12 in thick concrete columns as shown below. The beam is loaded at the center with a concentrated load of 50 kips. Idealizing the deep beam as a simple truss with concrete struts and bottom rebar ties, what is the approximate tension in the beam's bottom reinforcement? Effective depth of beam = 21 inches.



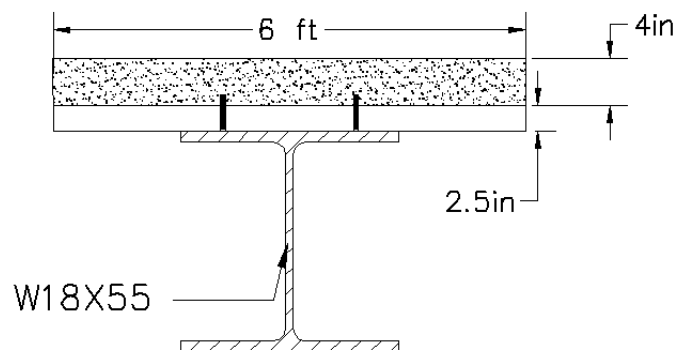
- (A) 25 kips
- (B) 50 kips
- (C) 75 kips
- (D) 100 kips

65. The critical elastic buckling load of a composite 16in X 16in concrete column with A992 Gr. 50 W12X72 steel core is 850 kips. What is the allowable compressive load on the composite section? $f'_c = 4000$ psi



- (A) 372 kips
- (B) 425 kips
- (C) 600 kips
- (D) 758 kips

66. A simply supported composite steel beam subjected to uniformly distributed load has two studs per rib as shown below. The total number of studs in the span is 40 and the nominal shear strength of each stud is 16.8 kips. Full composite action is not developed due to the limited number of studs. What is the moment of inertia of the effective composite section developed? Strength of concrete slab $f'_c = 5550$ psi



- (A) 1319 in⁴
- (B) 1631 in⁴
- (C) 1970 in⁴
- (D) 2980 in⁴

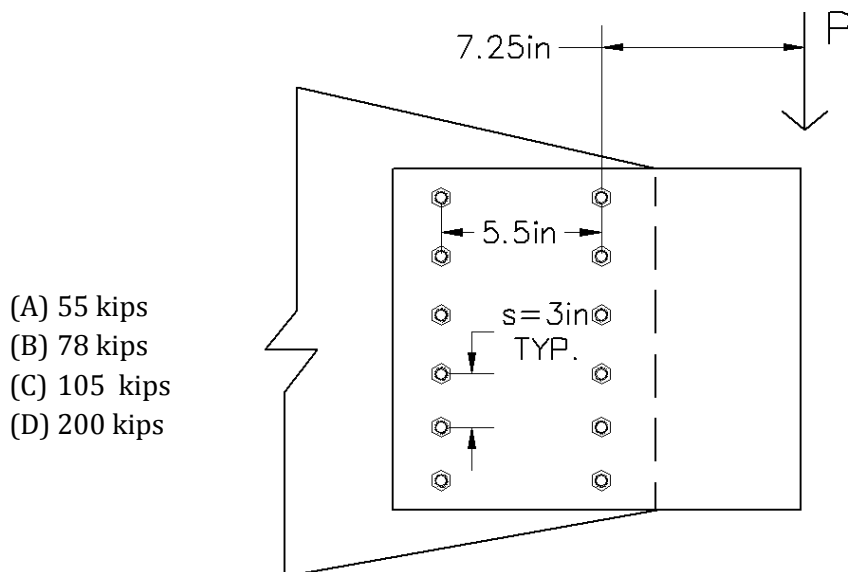
67. A special moment resisting frame joint has a 30 inX 30 in column and 20 in wide beams framing into it. According to ACI 318-14, which of the following is true about the transverse reinforcement in the joint?

- (A) Transverse reinforcement of the joint must match the transverse reinforcement of the columns above and below the joint,
- (B) The joint nominal shear strength is equal to $20\lambda\sqrt{f_c'}$ times joint area
- (C) No transverse moment is required if the column is fully in compression.
- (D) Spacing should not exceed 8 inches.

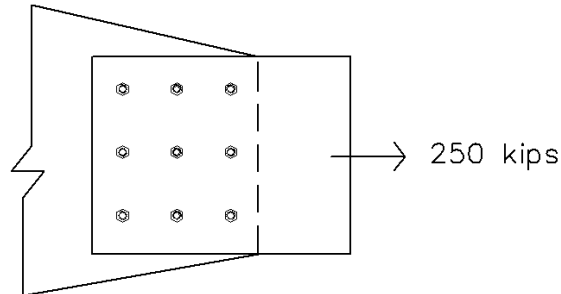
68. Which of the following is not true about restrained and unrestrained steel connections between a beam and a column?

- (A) Fully restrained connections are generally more simpler to analyze than partially restrained connections,
- (B) Partially restrained connections allow unrestrained relative rotation between the connected members and transmit moments partially,
- (C) Fully restrained connections allow negligible relative rotation between the connected members and transmit significant moments,
- (D) Fully restrained connections have high stiffness and partially restrained connections have intermediate level of stiffness

69. Using the instantaneous center of rotation method, what is the maximum allowable load "P" that the bolted connection shown below can withstand? The governing failure of the connection is bolt shear strength and allowable shear strength of each bolt is 20 kips.

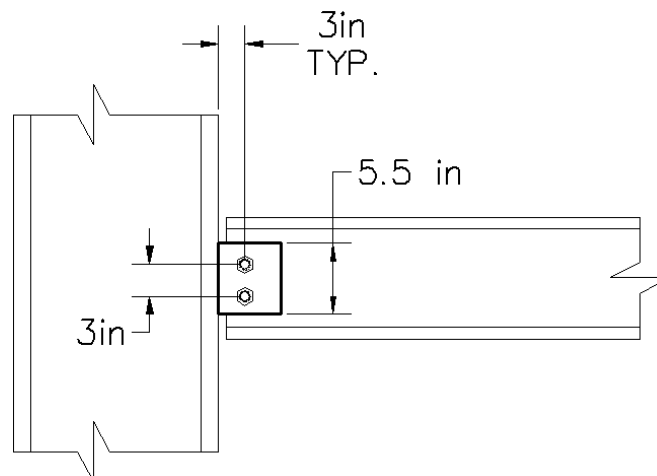


70. Two plates are connected as shown in the plan below and are subjected to a service load of 250 kips. The thickness of the top plate is $\frac{5}{8}$ in. The bolts are $\frac{3}{4}$ inch in diameter and the holes are standard size. What is the bearing stress on the bolts holes?



- (A) 26 ksi
- (B) 32 ksi
- (C) 47 ksi
- (D) 55 ksi

71. What is the allowable vertical reaction that can be transferred by the single plate shear connection shown below? The $\frac{3}{8}$ inch thick shear plate is sufficiently welded to the column and has sufficient edge distances per AISC requirements. The bolts used are $\frac{3}{4}$ inch diameter A325 and threads are not included in the shear plane. Plate $F_y = 36$ ksi.

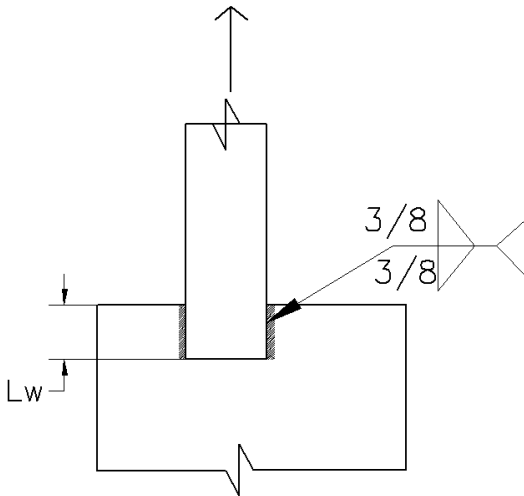


- (A) 21.1 kips
- (B) 25.3 kips
- (C) 27 kips
- (D) 29 kips

72. A pin connected hanger plate is connected to a support beam. The hanger plate is subjected to tension due to pick loads. The diameter of the pin is 3 inches and the thickness of the plate is 1.5 in. What is the allowable bearing strength of the connection?
 $F_y = 36$ ksi and $F_u = 58$ ksi for plates.

(A) 120.5 kips
(B) 124.4 kips
(C) 145.8 kips
(D) 194.4 kips

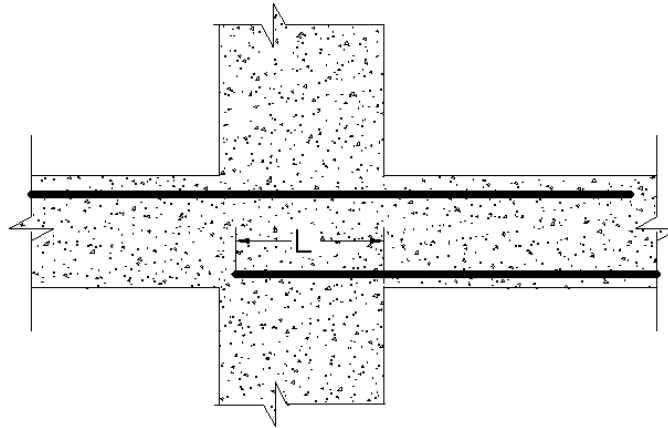
73. A plate is connected to a gusset by welding on two sides as shown below. The service tension load acting concentrically on the welds are 10 kips dead load and 25 kips live load. Assuming that the base metal failure does not govern, what is the required length of weld "L_w" ?
 $F_{exx} = 70$ ksi



(A) 2.75 in
(B) 3.25 in
(C) 3.5 in
(D) 4 in

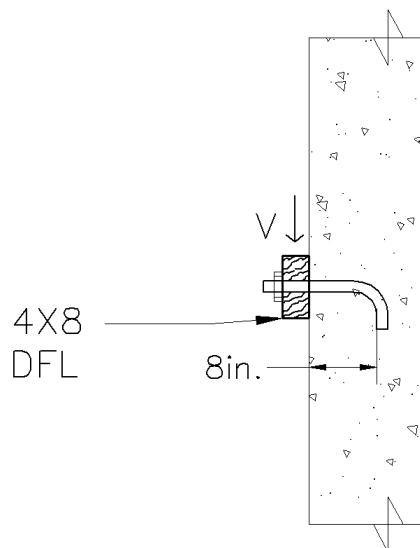
74. The beam on the right of the column shown below is a part of the primary lateral force resisting system. The bottom bars are #8 Gr. 60 deformed bars and the effective depth of the beam is 21 inches. Compressive strength of concrete is 4 ksi. There is no positive moment at the face of the column. The point of inflection is 2.5 ft away from the right face of the column. The clear cover is 1.5 in and the longitudinal bar clear spacing is 3 in. What is the minimum required extension 'L' of bottom bars from the face of the column? $\Psi_e = 1$, $\lambda = 1$

- (A) 0 in
(B) 6 in
(C) 21 in
(D) 48 in



75. A 1 inch diameter J bolt is used to anchor a 4X8 Douglas Fir-Larch ledger to a concrete shear wall as shown below. The bolt has a bending yield strength of 45000 psi and the concrete strength, bearing strength does not govern failure of the connection. What is the allowable vertical dead load on the ledger if the bolts are spaced at 4 ft c/c? Assume all adjustment factors except load factor are 1.

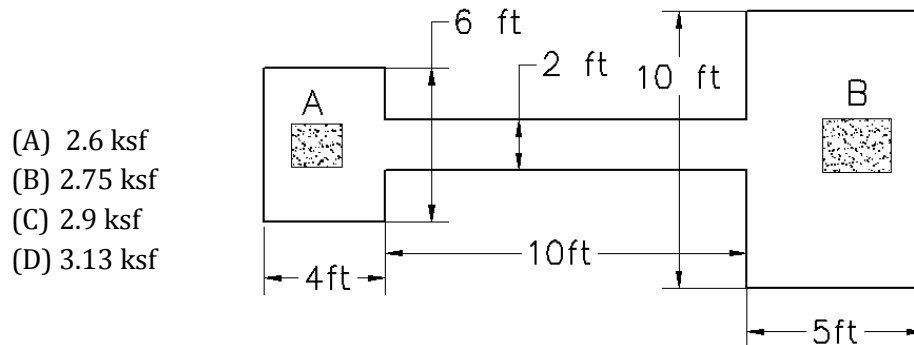
- (A) 165 plf
(B) 255 plf
(C) 300 plf
(D) 455 plf



76. A shallow strip footing is located 5 ft below the ground level. The footing is 3 ft wide. The density of soil is 115 pcf. The angle of internal friction of the soil is 25° and the cohesion = 300 psf. Calculate the ultimate bearing capacity for the footing. The factor of safety against bearing is 3.

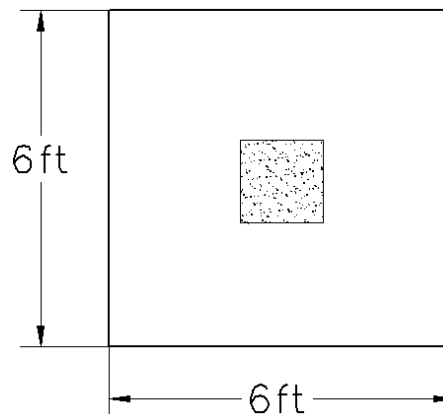
- (A) 3.25 ksf
- (B) 4.73 ksf
- (C) 8.28 ksf.
- (D) 14.24 ksf

77. The plan of a strap footing is shown below. The total compression load in column A is 58 kips and the total compression load in column B is 121 kips. The depth of the footings are 2.5 ft and the depth of strap beam is 2 ft. Bearing under strap beam is prevented by using a styrofoam layer underneath. What is most nearly the maximum bearing pressure under footing B? Unit weight of concrete = 150 pcf.



- (A) 2.6 ksf
- (B) 2.75 ksf
- (C) 2.9 ksf
- (D) 3.13 ksf

78. The plan of a square RCC footing is shown below. The dimensions of the column at the center of the footing is 18 in X 18 in. The factored concentric axial load acting in the column is 50 kips. What is the required design flexural strength of the footing?



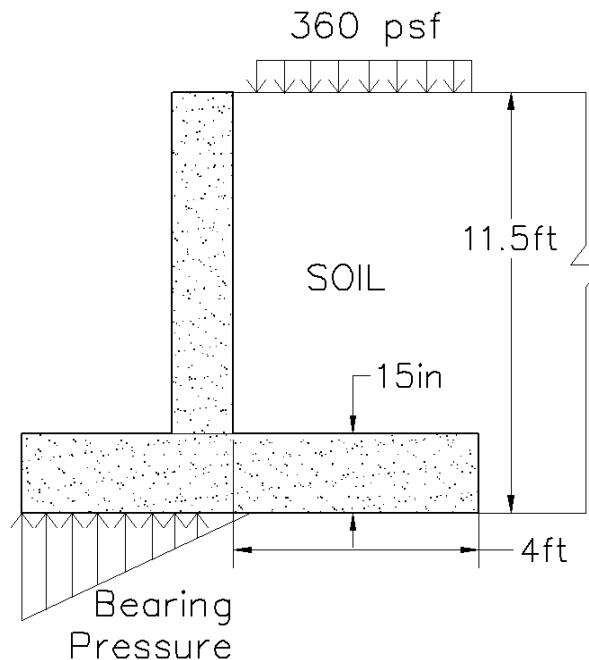
- (A) 12 kip*ft
- (B) 18 kip*ft
- (C) 21 kip*ft
- (D) 25 kip*ft

79. During the design of a pile group, when is it necessary to consider group effects ?

- (A) When the number of piles in a group is greater than four,
- (B) When the diameter of piles is greater than one third of the center to center spacing of the piles,
- (C) When the piles are end bearing,
- (D) When piles groups are driven in cohesive soil with significant scour depth.

80. A reinforced concrete cantilevered wall is retaining 11.5 ft tall backfill. A surcharge load of 360 psf acts on top of the backfill. The bearing pressure at the base of the wall due to combined eccentric loading is shown below. What is the maximum factored flexural demand on the heel? Consider equal load factor for soil and surcharge.
 Density of soil = 120 pcf Unit weight of concrete = 150 pcf

- (A) 11520 lb*ft/ft
- (B) 13226 lb*ft/ft
- (C) 18457 lb*ft/ft
- (D) 22152 lb*ft/ft



Solutions

[Click here for Problems](#)

1. Problem

Correct Answer: (D) 34 kip

Explanation:

Calculate the weight of water

Weight of 1 gallon = 8.34 lbf {Ref NCEES handbook page 2}

Weight of 2000 gallons = $2000 \times 8.34 = 16680$ lbf

Calculate weight of tank

Internal volume of tank = 2000 gallons

Solve for internal radius 'r' of tank

$$\frac{4}{3}\pi r^3 = 2000 * 0.134 \text{ cubic feet} \text{ {Ref NCEES handbook page 3 for conversion factors}}$$

$$r = 4 \text{ ft}$$

Calculate volume of 6 inches thick spherical concrete shell

$$\frac{4}{3}\pi(4 \text{ ft} + 6 \text{ inches})^3 - \frac{4}{3}\pi(4 \text{ ft})^3 = 113.62 \text{ cubic feet}$$

{Ref ACI 318-14 handbook page 34 for weight of normal weight concrete}

Weight of tank = $113.62 \text{ cubic feet} * 150 \text{ lb/ft}^3 = 17043$ lbf

Total dead load = Weight of water + Weight of tank

$$= 16680 \text{ lbf} + 17043 \text{ lbf} = 33723 \text{ lbf} \approx \mathbf{34 \text{ kips}}$$

2. Problem

Correct Answer: (C) 27 kips

Explanation:

{Ref ASCE 7-16 page 13, section 4.3.2 for provisions for partition}

Minimum partition wall load to be considered = 15 psf

Since the 1st floor is an open parking lot, the partition wall requirement does not apply.

Total floor area to be considered for partition walls = 30ftX30ft*2 floors

= 30ftx30ftx2

=1800 ft²

Total load = 15 psf*1800 ft²

=27000 lb

= **27 kips**

3. Problem

Correct Answer: (D) 2000 kips

Explanation:

{Ref ASCE 7-16 page 15, Table 4.3-1 for provisions for warehouse live loads}

Uniformly distributed warehouse load to be considered for heavy storage= 250 psf

Total area to be considered for storage = 50ftX100ft

=5000 ft²

Total load = 250 psf*5000 ft²

=1,250,000 lb

=1250 kips

Factored Live load = 1.6*1250 kips = **2000 kips**

4. Problem

Correct Answer: (B) 750 kips

Explanation:

Cross section area of HP 10X57 = 16.7 in^2 {Ref AISC Steel manual page 1-36}

Yield stress of material = 50 ksi

Allowable pile driving stress in H piles = $0.9 f_y$ {Ref NCEES PE handbook page 62}

= $0.9 * 50 \text{ ksi} = 45 \text{ ksi}$

Allowable pile driving load = Allowable driving stress * cross section area

= $45 \text{ ksi} * 16.7 \text{ in}^2$

= 751.5 kips

≈ **750 kips**

5. Problem

Correct Answer: (A)

Explanation:

Per ASCE 7-16 page 246,

For analysis under wind loads, diaphragms constructed of untopped steel decks, concrete-filled steel decks, and concrete slabs, each having a span-to-depth ratio of 2 or less, shall be permitted to be idealized as rigid.

The span to depth in the direction of wind for floor A is $50 \text{ ft} / 20 \text{ ft} = 2.5$, exceeds 2. Thus it cannot be idealized as rigid although it is a concrete slab.

6. Problem

Correct Answer: (B) 2 kips

Explanation:

The plan shows two walls along the length of the diaphragm which will have high in-plane stiffness compared to the small central columns. Hence, the rigid diaphragm will transfer the loads equally to two walls on two sides with similar dimensions and stiffness.

Load shared by each wall = $(200 \text{ lb/ft} \times 20 \text{ ft}) / 2 = 2000 \text{ lb} = \mathbf{2 \text{ kip}}$

7. Problem

Correct Answer: (D) 25% of floor live load for slab on grade ground floor storage

Explanation:

Per ASCE 7-16 section 12.7.2, effective seismic weight, options A, B and C are all correct. If we look at option D, it says 25% of floor live load for areas used for storage (which is true per section 12.7.2). However, option D also says that the storage area is a ground floor slab on grade – meaning all the loads from that floor is directly transferred to the bearing soil via slab on grade independent of framing elements of the building. Hence it can be excluded from seismic weight calculation and the base shear can be reduced.

8. Problem

Correct Answer: (B) 250 kip*ft

Explanation:

The wheel will act like a point load and will cause maximum bending moment at the beam when it is at the center of the span.

Ref Beam formulas from NCEES PE handbook page 238

$P = 50 \text{ kips}$ $L = 20 \text{ ft}$

$M = P \cdot L / 4 = 50 \text{ kips} \cdot 20 \text{ ft} / 4 = \mathbf{250 \text{ kip} \cdot \text{ft}}$

9. Problem

Correct Answer: (B) 17 psf

Explanation:

From ASCE 7-16, Table 1.5-2, page 5, for risk category II,
snow importance factor $I_s = 1$

From page 58, Table 7.3-1, for surface roughness C and fully exposed roof,
exposure factor $C_e = 0.9$

From page 58, Table 7.3-2, for well heated roof,
thermal factor $C_t = 1$

Ground snow load, $p_g = 17 \text{ psf}$

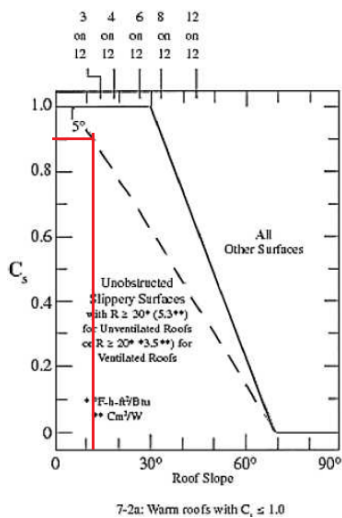
From equation 7.3-1 page 51, flat roof snow load

$$p_f = 0.7 * C_e * C_t * I_s * p_g$$

$$p_f = 0.7 * 0.9 * 1 * 1 * 17 \text{ psf} = 10.7 \text{ psf}$$

$$\text{Slope of roof} = \tan^{-1} (6\text{ft}/25\text{ft}) = 13.5 \text{ degrees}$$

From page 59, from 1st graph of figure 7.4-1, slope factor with $C_t = 1$, slope = 13.5 degrees
and slippery surface



$$C_s = 0.9$$

From equation 7.4-1

$$P_s = C_s * p_f = 0.9 * 10.7 = 9.63 \text{ psf}$$

Check criteria for minimum snow load from section 7.3.4, monoslope roof for ground snow
load less than 20 psf

$$P_m = I_s * p_g = 1 * 17 \text{ psf} = 17 \text{ psf}$$

Hence minimum load requirement governs, and correct answer is **17psf**

10. Problem

Correct Answer: (C) 10 kips

Explanation:

Per AASHTO LRFD 2017 section 3.6.4, page 3-33, the braking force is taken as the greater of 25% of the weights of design truck/tandem or 5% of the sum of lane load and truck/tandem.

Here Reaction due to truck = 40 kips

25% of 40 kips = 10 kips

Sum of reactions due to truck and lane load

= 40 kips + 65 kips

= 105 kips

5% of 105 kips = 5.25 kips

Thus, 'truck only' governs and estimated braking force = **10 kips**

11. Problem

Correct Answer: (B)

Explanation:

All options are incorrect except option B.

Sands and gravels generally do not swell and clayey soils are more susceptible. Per ASCE 7-16, page 12, section 3.2.1, The lateral pressure shall be increased if expansive soils are present at the site, as determined by a geotechnical investigation.

12. Problem

Correct Answer: (B) 11.38 lb/ft²

Explanation:

Lateral pressure intensity due to infinitely long line load is given by AASHTO LRFD 2017 page 3-137 equation 3.11.6.2-3

Here, X = 5ft since load is acting 5ft away from the wall

Z = 4ft, depth at which intensity is required

Q = 150 lb/ft

Calculate R

$$R^2 = X^2 + Z^2$$

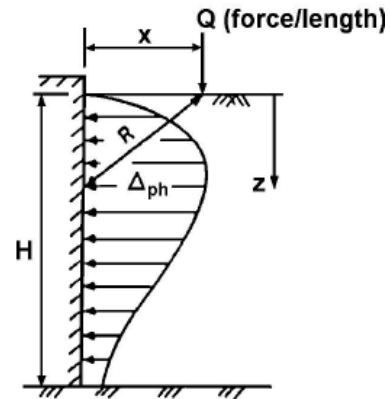
$$R^2 = 5^2 + 4^2$$

$$R = 6.4\text{ft}$$

At depth Z, the intensity of pressure is given by

$$\Delta_{ph} = \frac{4 \cdot Q}{\pi} * \frac{X^2 \cdot Z}{R^4}$$

$$\Delta_{ph} = \frac{4 \cdot 150\text{lb/ft}}{\pi} * \frac{(5\text{ft})^2 \cdot 4\text{ft}}{(6.4\text{ft})^4} = 11.38\text{ lb/ft}^2$$



13. Problem

Correct Answer: (C) 833.33 lbs

Explanation:

Diaphragms act as beams and the chord forces are the tension and compression forces that act in the outer boundaries of a diaphragm.

The diaphragm is analyzed as simply supported beam with Uniform loading of 200 lb/ft

Loading, W = 200 lb/ft

Depth, d = 12ft

Span, L = 20ft

$$\text{Moment} = W \cdot L^2 / 8$$

$$= \{200\text{ lb/ft} * (20\text{ft})^2\} / 8$$

$$= 10,000 \text{ lb}\cdot\text{ft}$$

$$\begin{aligned}\text{Chord force} &= M/d \\ &= \{10,000 \text{ lb}\cdot\text{ft}\}/12\text{ft} \\ &= \mathbf{833.33 \text{ lbs}}\end{aligned}$$

14. Problem

Correct Answer: (C) Typically, 'Strength IV' combination governs over 'Strength I' if the dead load effects are at least two times the live load effects,

Explanation:

All other options are correct per AASHTO LRFD section 3.4 except option C. Per page 3-9, typically Strength IV governs if the dead load to live load ratio is above 7.

15. Problem

Correct Answer: (D)

Explanation:

The section with the highest moment of inertia about Y-Y will have the highest flexural stiffness. From the given options, Option C will have the lowest moment of inertia about YY axis. Options A and B have higher moment of inertias than C but less than Option D, Hence D is the correct answer.

16. Problem

Correct Answer: (D)

Explanation:

Since there is a hinge at the center of the beam, Option A and option B are ruled out because they don't show sufficient rotation about the hinge.

Option C shows a case when the uniformly distributed load would be acting in upwards direction. Hence the correct answer is Option D.

17. Problem

Correct Answer: (B)

Explanation:

All options are correct except option B. Influence lines describe fundamental properties of the structure and the shape of the diagram is independent of the system of units used.

18. Problem

Correct Answer: (D) Cannot be determined

Explanation:

Ref to NCEES PE handbook page 34 for critical buckling load

The column is free on one side and pinned on the other side. For pinned=free condition, the effective length factor 'K' is very huge and indeterminate as the configuration itself is unstable. This makes the value of critical buckling load close to zero. Hence the best answer for this question is option D 'Cannot be determined'

19. Problem

Correct Answer: (B) 170.7 kips

Explanation:

Ref to NCEES PE handbook page 34 for critical buckling load and page 25 for section properties.

$K = 2$ for fixed - free condition

$L = 20\text{ft} = 240\text{ inches}$

Calculate moment of inertia of the section

$a = 4\text{ inches}$

$b = a - \text{wall thickness} = 4\text{ inches} - 1\text{ inches} = 3\text{ inches}$

$$I = \pi * (a^4 - b^4)/4$$

$$I = \pi * (4^4 - 3^4)/4 = 137.44\text{ in}^4$$

Calculate critical axial buckling load

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} = \frac{\pi^2 * 29000 * 137.44}{(2 * 240)^2} = 170.7\text{ kips}$$

Correct answer is option B.

20. Problem

Correct Answer: (D)

Explanation:

Ref to NCEES PE handbook page 34 for critical buckling load.

The column self weight is assumed insignificant to buckling load and is neglected.

The material of the column is assumed homogeneous, isotropic and elastic, obeying Hooke's law.

Deformation including axial shortening of columns due to compression is disregarded.

Column is assumed to be subjected to purely axial load only.

Hence all options are incorrect except option D.

21. Problem

Correct Answer: (D) 6

Explanation:

When we analyze joint A and E, there is no external load. If there are ONLY two non-collinear members at a joint, with no external load, both are zero force members. Hence AB, AI, ED and EF are zero force members.

When we analyze joint C, there is no external load. If there are ONLY three members at a joint, two of which are collinear, with no external load, then the non collinear member is zero force member. So CH is also a zero force member.

Since AI is a zero force member, IH is also zero because force in AI = IH.

Hence AB, AI, ED, EF, CH and IH are zero force members and the correct answer is option D.

22. Problem

Correct Answer: (B) 16.5 kips

Explanation:

Ref to NCEES PE handbook page 33 for transverse shear stress flow in beams

Calculate location of neutral axis

Location of centroid of section from base,

$$y_w = (0.5\text{ in} + 0.5\text{ in} + 2\text{ in} + 0.5\text{ in} + 0.5\text{ in})/2 = 2\text{ in (due to symmetry)}$$

Calculate total moment of inertia about centroid

$$\begin{aligned} I_{xx} &= \{4\text{ in} * (4\text{ in})^3\}/12 - \{3.5\text{ in} * (2\text{ in})^3\}/12 \\ &= 19\text{ in}^4 \end{aligned}$$

Calculate the first moment of area above the junction of top flange plates

i.e. at depth 0.5 in from the top of section,

$$Q = 4\text{ in} * 0.5\text{ in} * (2\text{ in} - 0.25\text{ in}) = 3.5\text{ in}^3$$

Shear force, $V = 15\text{ kips} = 15000\text{ lbs}$

Transverse shear flow per unit length of beam along the span,

$$\begin{aligned} \tau &= (VQ)/I \\ &= (15000\text{ lbs} * 3.5\text{ in}^3)/(19\text{ in}^4) \\ &= 2763.15\text{ lbs/in} \end{aligned}$$

Spacing of bolts, $s = 12\text{ inches}$

There are 2 bolts, one on either side of the flange sharing this shear

Thus shear per bolt

$$\begin{aligned} &= (\tau * s)/2 \\ &= (2763.15\text{ lbs/in} * 12\text{ in})/2 \\ &= 16578.94\text{ lbs} \end{aligned}$$

$\approx 16.5\text{ kips}$

Hence, correct answer is option B

23. Problem

Correct Answer: (A) 21.23 ksi

Explanation:

The 1st (top) and the 2nd plate connection has only 1 shear plane.

The 2nd, 3rd and 4th plates have two shear planes.

Since both P1 and P2 are equal, shear stress will be maximum in the bolts under the top plate in which 6 bolts are under single shear.

No. of bolts, $n = 6$

$P1 = 100$ kips

Area of 1 inch bolts

$$= \pi/4 * (1 \text{ in})^2$$

$$= 0.785 \text{ in}^2$$

$$\text{Maximum nominal shear stress} = 100 \text{ kips} / (6 * 0.785 \text{ in}^2) = \mathbf{21.23 \text{ ksi}}$$

Correct answer is option A.

24. Problem

Correct Answer: (B)

Explanation:

Increasing the depth of the beam section will not reduce the shear force.

Reducing the span b will reduce the vertical reaction at fixed support and will also reduce the shear at that location.

Shear is directly proportional to the magnitude of the load so reducing load will reduce the shear. Hence correct answer is option B.

25. Problem

Correct Answer: (D) 3830 kip*ft

Explanation:

Since all story height are equal

Story height = 40 ft/4 = 10ft

Lever arm for 55 kips load = 40ft

Lever arm for 37 kips load = 30ft

Lever arm for 21 kips load = 20ft

Lever arm for 10 kips load = 10ft

Total overturning moment

= 55 kip*40 ft+37 kip*30 ft+21 kip*20 ft+10 kip*10 ft

= **3830 kip*ft**

Correct answer is option D

26. Problem

Correct Answer: (A) 17.4 ft

Explanation:

Maximum bending moment is located where the shear force is zero.

Calculate reaction Rb

Sum of moments about A = 0

$R_b * 30\text{ft} = 30 \text{ kips} * 18 \text{ ft} + 5 \text{ kips/ft} * (30\text{ft})^2/2$

$R_b = 93 \text{ kips}$

Sum of vertical forces = 0

$R_a = 30 \text{ kips} + 5 \text{ kip/ft} * 30\text{ft} - R_b$

$R_a = 87 \text{ kips}$

Equation of shear force from left end is:

At any point x, (x = 0 to 18ft)

$V_x = R_a - 5 \text{ kip/ft} * x$

At x = 0, $V_x = 87 \text{ kips}$

At x = 18 ft, $V_x = 87 \text{ kips} - 5 \text{ kip/ft} * 18 = -3 \text{ kips}$

Hence point of zero force lies between 0 ft and 18 ft

If $V_x = 0$

$0 = R_a - 5 \text{ kip/ft} * x$

$$x = 87 \text{ kip} / (5 \text{ kip/ft}) = \mathbf{17.4 \text{ ft}}$$

Correct answer is option A.

27. Problem

Correct Answer: (B) 15.5 ksi compressive, 30.5 ksi tensile

Explanation:

Ref to NCEES PE handbook page 29 for principal stresses.

$$\sigma_x = 25 \text{ ksi (tensile stress)}$$

$$\sigma_y = -10 \text{ ksi (compressive stress)}$$

$$\tau_{xy} = 15 \text{ ksi (shear stress)}$$

$$\sigma_a = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_a = \frac{25 - 10}{2} + \sqrt{\left(\frac{25 - (-10)}{2}\right)^2 + 15^2} = 30.5 \text{ ksi (tensile)}$$

$$\sigma_b = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_b = \frac{25 - 10}{2} - \sqrt{\left(\frac{25 - (-10)}{2}\right)^2 + 15^2} = -15.5 \text{ ksi (compressive)}$$

Correct answer is option B.

28. Problem

Correct Answer: (C) 8 ksi

Explanation:

$$\text{Area of section, } A = 6 \text{ in} \times 10 \text{ in} = 60 \text{ in}^2$$

$$\text{Eccentricity, } e = 10 \text{ in} / 2 = 5 \text{ in}$$

$$\text{Load, } P = 120 \text{ kip}$$

$$\text{Section modulus about bending axis, } S = 6 \text{ in} \times 10 \text{ in} \times 10 \text{ in} / 6$$

$$S = 100 \text{ in}^3$$

$$\text{Moment, } M = P \times e$$

$$= 120 \text{ kip} \times 5 \text{ in}$$

$$= 600 \text{ kip} \cdot \text{in}$$

Maximum compressive stress

$$= P/A + M/S$$

$$= 120 \text{ kip} / 60 \text{ in}^2 + 600 \text{ kip}\cdot\text{in} / 100 \text{ in}^3$$

$$= \mathbf{8 \text{ ksi}}$$

Correct answer is option C

29. Problem

Correct Answer: (D)

Explanation:

Shear forces are forces that act parallel to the cross section of the body i.e. tangentially to the surface of the body. Unlike point loads in a cantilever, this does not produce bending and the body is purely in shear. Hence, the correct answer is option D 'shear only'.

30. Problem

Correct Answer: (C) **1.3 kip/ft**

Explanation:

Ref to NCEES PE handbook pages 233 and 237 for deflections due to uniform loadings

$$I = 60 \text{ in}^4$$

$$E = 29000 \text{ ksi}$$

$$\text{span, } l = 20 \text{ ft} = 240 \text{ in}$$

$$\text{Total loading due to uniformly varying load, } W = 0.5 \cdot 20 \text{ ft} \cdot 2 \text{ kips/ft} = 20 \text{ kips}$$

Maximum deflection is given by

$$\Delta_{\text{max}} = W l^3 / (60EI)$$

$$= 20 \text{ kips} \cdot (240 \text{ in})^3 / (60 \cdot 29000 \text{ ksi} \cdot 60 \text{ in}^4) = 2.65 \text{ in}$$

Deflection caused by a uniformly distributed loading of intensity w is given by

$$\Delta_{\text{max}} = 5w l^4 / (384EI)$$

$$w = \Delta_{\text{max}} \cdot EI / l^4 \cdot (384/5)$$

$$w = 2.65 \text{ in} \cdot 29000 \text{ ksi} \cdot 60 \text{ in}^4 / (240 \text{ in})^4 \cdot (384/5)$$

$$w = 0.107 \text{ kip/in} = 1.28 \text{ kip/ft}$$

$$\approx \mathbf{1.3 \text{ kip/ft}}$$

Correct answer is option C.

31. Problem

Correct Answer: (D) One sixteenth

Explanation:

Ref NCEES PE handbook page 236 for beam deflections.

When depth of beam is doubled, the new moment of inertia is

$$I_{\text{new}} = b(2h)^3/12 = 8(bh^3/12) = 8 * I_{\text{original}}$$

$$\text{Loadings, } w_{\text{new}} = (1/2) w_{\text{original}}$$

Deflection is directly proportional to loading 'w' and inversely proportional to 'I'

$$\Delta_{\text{new}} / \Delta_{\text{original}} = w_{\text{new}} / w_{\text{original}} * I_{\text{original}} / I_{\text{new}}$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = \{(1/2) w_{\text{original}}\} / w_{\text{original}} * I_{\text{original}} / (8 * I_{\text{original}})$$

$$\Delta_{\text{new}} / \Delta_{\text{original}} = (1/2) * (1/8) = \mathbf{1/16.} \quad \text{Hence the correct answer is option D.}$$

32. Problem

Correct Answer: (B)

Explanation:

Modulus of elasticity of gray cast iron is less than mild steel and the deflections are inversely proportional to the modulus of elasticity. Hence, mild steel with higher 'E' will deflect less than gray cast iron beam for identical conditions. All other options are incorrect and the correct answer is option B.

33. Problem

Correct Answer: (C)

Explanation:

Lateral torsional buckling occurs when beams are subjected to transverse loads (or combination of transverse and axial loads) and have long unsupported spans. Hence, the correct answer is option C.

34. Problem

Correct Answer: (A) 3.77 ksi

Explanation:

Ref AASHTO LRFD bridge design specifications page 6-50, section 6.6.1.2.5 for fatigue resistance.

AADT = 2000

From AASHTO LRFD Table 6.6.1.2.3-1 page 6-39. This detail belongs to Category C'.

Value of constant A = 44×10^8 ksi

From table 6.6.1.2.5-2 page 6-52,

Cycles per truck passage, for continuous girders , near interior support,

$n = 1.5$

From equation 6.6.1.2.5-3, page 6-50

$N = 365 \times 75 \times n \times \text{AADT}$

$N = 365 \times 75 \times 1.5 \times 2000$

$N = 82125000$

From equation 6.6.1.2.5-2, page 6-50

Fatigue resistance = $(A/N)^{1/3}$

$= (4400000000 / 82125000)^{1/3}$

= 3.77 ksi

Correct answer is option A

35. Problem

Correct Answer: (B)

Explanation:

The department of building official is the one who approves procedures, submittals and permits related to special inspection and cannot be the special inspector. A professional employed by the contractor and the subcontractor are not independent from the contractor and cannot work as a special inspector for that project. The correct answer is the engineer of record of the building who can work as both the design professional and a special inspector if they have the necessary qualifications and license/registration.

36. Problem

Correct Answer: (B)

Explanation:

Contractors do not approve submittal documents or revise any design assumptions. The engineer of record will be notified by the department of building directly when the design drawings are approved because it is usually the responsibility of the EOR to get the design approved. Hence the correct answer is option B, to compile and review all submittals before forwarding it to the building engineer of record, so that the EOR can approve it.

37. Problem

Correct Answer: (C) 1/2 inch

Explanation:

Tension load in the anchor, $T = 2.5 \text{ kips} * \sin(45 \text{ degree}) = 1.768 \text{ kips}$

Shear load in the anchor, $V = 2.5 \text{ kips} * \cos(45 \text{ degree}) = 1.768 \text{ kips}$

Factor of safety = 2

Check with 3/8 inch anchor

Allowable forces = Ultimate forces / Factor of safety

Tension allowable, $T_a = 5.94 \text{ kips} / 2 = 2.97 \text{ kips}$

Shear allowable, $V_a = 4.14 \text{ kips} / 2 = 2.07 \text{ kips}$

Hence 3/8 inch anchor satisfies individual shear tension requirement.

Check interaction since both forces are acting together

$$(T / T_a) + (V / V_a)$$

$$= 1.768 / 2.97 + 1.768 / 2.07 = 1.44$$

This should be less than or equal to 1 for the anchor to be safe. Hence it doesn't work

Check 1/2 inch anchor

Tension allowable, $T_a = 9.64 \text{ kips} / 2 = 4.82 \text{ kips}$

Shear allowable, $V_a = 7.24 \text{ kips} / 2 = 3.62 \text{ kips}$

$$(T / T_a) + (V / V_a)$$

$$= 1.768 / 4.82 + 1.768 / 3.62 = 0.855, \text{ Safe}$$

Hence, the correct answer is Option C, **1/2 inch**.

38. Problem

Correct Answer: (D) 7.6 kips

Explanation:

Dead Load due to Slab weight = $150 \text{ pcf} \times 0.5 \text{ ft}$

$$= 75 \text{ psf}$$

Total load from 6th floor at an area 5 ft X 5 ft

$$= (75 \text{ psf} + 75 \text{ psf} + 15 \text{ psf}) \times 5 \text{ ft} \times 5 \text{ ft}$$

$$= 4125 \text{ lbs}$$

$$= 4.125 \text{ kips}$$

Total load from 5th floor at an area 5 ft X 5 ft

$$= (75 \text{ psf} + 50 \text{ psf} + 15 \text{ psf}) \times 5 \text{ ft} \times 5 \text{ ft}$$

$$= 3500 \text{ lbs}$$

$$= 3.5 \text{ kips}$$

Total load on shoring post

$$= 4.125 \text{ kips} + 3.5 \text{ kips}$$

$$= 7.625 \text{ kips} \approx \mathbf{7.6 \text{ kips}}$$

Option D is the correct answer.

39. Problem

Correct Answer: (B) 4 kips

Explanation:

Spacing of vertical posts = 30 in c/c

Spacing of horizontal wales = 24 inches c/c

Since every alternate post is tied to the wales,

Tributary area for each tie point

$$= (30 \text{ in} \times 2) \times 24 \text{ in}$$

$$= 1440 \text{ in}^2$$

$$= 10 \text{ ft}^2$$

Lateral pressure = 400 psf

Load on each tie = 400 psf * 10 ft²

$$= \mathbf{4 \text{ kips}}$$

Correct answer is Option B.

40. Problem

Correct Answer: (D) 200%

Explanation:

Ref to NCEES PE handbook page 70, section 2.6.2

For 90 dBA, time permitted = 8 hours, time spent = 4 hours

For 100 dbA, time permitted = 2 hours, time spent = 2 hours

For 95 dbA, time permitted = 4 hours, time spent = 2 hours

Noise dose, $D = 100 \times \sum (\text{time spent at specified sound pressure} / \text{time permitted})$

$$D = 100 \times (4/8 + 2/2 + 2/4)$$

$$D = \mathbf{200\%}$$

Correct answer is Option D.

41. Problem

Correct Answer: (B) I - Gap graded II- Well graded III- Uniform

Explanation:

Sample I shows a non uniform irregular curve which denotes that one or more ranges of particle sizes are missing. Hence it is gap graded.

Sample II shows a continuous curve spreading over a wide range of particle size which denotes a well graded sample. Sample III shows a narrow vertical curve which denotes uniformity in particle size and hence is uniformly graded.

Correct answer is option B.

42. Problem

Correct Answer: (D) 131.6 pcf

Explanation:

Ref to NCEES PE handbook page 285, section 5.5.1,

$$\text{Relative density, } D_R = \frac{e_{max} - e}{e_{max} - e_{min}}$$

e = in situ void ratio of soil

e.max and e.min are void ratios in loosest and densest conditions

From PE handbook page 126, chart showing correlation between SPT resistance and D_R ,

For overburden of 4 ksf and $N = 20$, relative density $\approx 60\%$

Then,

$$60\% = \frac{e_{max} - e}{e_{max} - e_{min}} = \frac{0.4 - e}{0.4 - 0.2}$$

Solve for 'e',

$$e = 0.28$$

$\gamma_w = 62.4$ pcf, unit weight of water from PE handbook page 2,

$$G_s = 2.7$$

From PE handbook page 133

$$\gamma_d = \frac{G_s * \gamma_w}{1 + e}$$

$$\gamma_d = \frac{2.7 * 62.4 \text{ pcf}}{1 + 0.28}$$

$$= 131.6 \text{ pcf}$$

Correct answer is option D

43. Problem

Correct Answer: (D) 4

Explanation:

Ref to ACI 318-14, page 480 Section 26.12.3.1(b),

For compliance, the arithmetic average of any three consecutive strength tests equals or exceeds f_c' and no strength test falls below f_c' by more than 500 psi if f_c' is 5000 psi or less.

Option A is not true because Day 1 and Day 2 results although less than 3500 psi is not less than $f_c' - 500 \text{ psi} = 3500 \text{ psi} - 500 \text{ psi} = 3000 \text{ psi}$

Option B and C is not true because the mix fails because of results from day 5, the results are less than 3000 psi.

Hence Option D is true and the answer.

44. Problem

Correct Answer: (D) 0.9

Explanation:

From NCEES PE handbook page 260, area of #6 bars = 0.44 in^2

$$A_s = 5 \times 0.44 \text{ in}^2 = 2.2 \text{ in}^2$$

Calculate depth of stress block 'a' from equation in page 264 section 4.3.2.2

$$a = A_s \cdot f_y / (0.85 \cdot f'_c \cdot b)$$

$$a = 2.2 \cdot 60000 / (0.85 \cdot 4000 \cdot 12)$$

$$a = 3.24 \text{ in}$$

Ref to NCEES PE handbook, page 263, section 4.3.2.1

Ratio of depth of stress block "a" to neutral axis depth "c"

$\beta_1 = 0.85$ for 4000 psi concrete

$$a/c = 0.85$$

$$c = a / 0.85 = 3.81 \text{ in}$$

Strain in concrete = 0.003

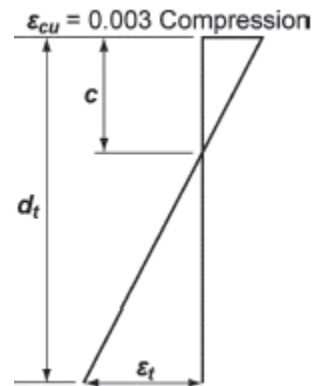
$$\text{Strain in rebar} = 0.003 \cdot (d - c) / c$$

$$= 0.003 \cdot (20 - 3.81) / 3.81$$

$$= 0.0127$$

From ACI 318-14, page 343, table 21.2.2, section is tension controlled and $\Phi = 0.9$

Correct answer is option D.



45. Problem

Correct Answer: (C) 4494 in⁴

Explanation:

From NCEES PE handbook page 260, area of #6 bars = 0.44 in²

$$A_s = 4 \times 0.44 \text{ in}^2 = 1.76 \text{ in}^2$$

$$\text{Modular ratio, } n = 29000/3600 = 8.06$$

Calculate depth of N.A. from equation in page 265, section 4.3.2.3

$$p = A_s/bd = 1.76 \text{ in}^2 / (12 \text{ in} \times 22 \text{ in}) = 0.0067$$

$$k = \sqrt{[(p \cdot n)^2 + 2p \cdot n]} - p \cdot n$$

$$k = \sqrt{[(0.0067 \cdot 8.06)^2 + 2 \cdot 0.0067 \cdot 8.06]} - 0.0067 \cdot 8.06$$

$$k = 0.279$$

Depth of neutral axis

$$c = kd$$

$$c = 0.279 \times 22 \text{ in}$$

$$c = 6.138 \text{ in}$$

Moment of inertia of the cracked transformed section about the N.A. is given by

Sum of transformed rebar area $n \cdot A_s$ and depth of concrete block $c \cdot b$ about the base

MOI of rectangular block about the base

$$= 1/3 \cdot b \cdot c^3$$

$$= 1/3 \cdot 12 \text{ in} \cdot (6.138 \text{ in})^3$$

$$= 924.99 \text{ in}^4$$

MOI of transformed rebar area about NA

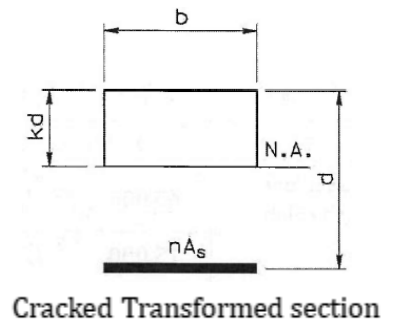
$$= n \cdot A_s \cdot (d - c)^2$$

$$= 8.06 \cdot 1.76 \text{ in}^2 \cdot (22 \text{ in} - 6.138 \text{ in})^2$$

$$= 3569.14 \text{ in}^4$$

$$\text{Total MOI} = 924.99 \text{ in}^4 + 3569 \text{ in}^4 = 4494.13 \text{ in}^4$$

Hence the correct answer is Option C **4494 in⁴**.



46. Problem

Correct Answer: (B) 12 ksi

Explanation:

Ref to AISC steel manual 15th edition page 16.1-129, AISC 360-16 Table J3.2

For A307 bolts,

Nominal tensile strength, $F_{nt} = 45 \text{ ksi}$

Nominal shear strength, $F_{nv} = 27 \text{ ksi}$

From page 16.1-133, section J3.7 equation J3-3b,

Modified nominal tensile strength to include effects of shear

$$= 1.3 * F_{nt} - \Omega * F_{nt} / F_{nv} * f_{rv}$$

$$= 1.3 * 45 \text{ ksi} - 2 * 45 \text{ ksi} / 27 \text{ ksi} * 10 \text{ ksi}$$

$$= 25.167 \text{ ksi}$$

Allowable tensile stress

$$= 25.167 \text{ ksi} / \Omega$$

$$= 25.167 \text{ ksi} / 2$$

$$= 12.58 \text{ ksi}$$

Correct answer is option B, **12 ksi**.

47. Problem

Correct Answer: (B) 8.35 kip/in

Explanation:

Ref to AISC steel manual 15th edition page 8-8 for available strength of fillet welds,

For 3/8 in weld,

Size of weld in 16th of an inch

$$\text{Size} = (3*2)/(8*2) \text{ in} = 6/16 \text{ in, Thus } D = 6$$

$$F_{exx} = 70 \text{ ksi}$$

From equation 8-1,

Nominal strength per unit length

$$R_n = 0.6 * F_{exx} * (1/\sqrt{2}) * (D/16)$$

$$R_n = 0.6 * 70 * (1/\sqrt{2}) * (6/16)$$

$$R_n = 11.137 \text{ kip/in}$$

$$\text{Allowable Strength} = R_n / \Omega = 11.137 / 2 = 5.57 \text{ kip/in}$$

From page 8-9, effect of loading angle on strength, when angle = 90 degrees for transversely loaded welds, increase in strength of 50% is applicable

Thus , Allowable strength = 5.57 kip/in * 1.5 = **8.35 kip/in**

Correct answer is option B.

48. Problem

Correct Answer: (D)

Explanation:

All options are true except Option D.

Dimensional tolerances for cold formed steel are tighter than that for hot rolled sections.

This is because cold working allows for more precise control over the dimensions. Hence

Option D is false and the answer.

49. Problem

Correct Answer: (A) **31 in⁴**

Explanation:

Ref to NDS 2018 Supplement page 34 Table 4A.

For select structural DFL, E = 1900000 psi

For dry conditions, C_m = 1

From NDS 2018 page 11, Table 2.3.3, for <100°F, C_t = 1

From NDS 2018 page 29, Table 4.3.1

$$E' = E * C_m * C_t * C_i$$

$$E' = 1900000 \text{ psi} * 1 * 1$$

$$E' = 1900000 \text{ psi}$$

Per IBC 2018 page 363 Table 1604.3, for live loads in floor members,

$$\Delta = L/360$$

$$\Delta = (12\text{ft} * 12 \text{ in/ft}) / 360 = 0.4 \text{ in}$$

$$w = 50 \text{ plf}$$

$$w = 50/12 \text{ lbs/in} = 4.167 \text{ lbs/in}$$

From NCEES handbook page 236 Case I,

$$\Delta = 5 * w * L^4 / (384 EI)$$

$$I = 5 * w * L^4 / (384 E \Delta)$$

$$I = [5 * 4.167 \text{ lb/in} * (144\text{in})^4] / (384 * 1900000 \text{ psi} * 0.4 \text{ in})$$

$$I = 30.7 \text{ in}^4 \quad \text{Correct answer is Option A, } \mathbf{31 \text{ in}^4}.$$

50. Problem

Correct Answer: (D) 65 psi

Explanation:

Ref to TMS 402/602 -16 page C-110 Table 8.2.4.2

Allowable flexural tensile stresses for clay and concrete masonry

For hollow, fully grouted unit and Type M mortar,

Allowable stress = **65 psi**

Correct answer is Option D.

51. Problem

Correct Answer: (C) 2.75 hour

Explanation:

Ref to IBC 2018 page 189, Table 722.3.2

For concrete masonry constructed with calcareous gravel and equivalent thickness 5 inches,

Fire resistance rating = **2.75 hour**

Correct answer is Option C.

52. Problem

Correct Answer: (A) Toughness test

Explanation:

The given procedure records energy required to cause failure of the specimen due to impact from the pendulum. This is a measure of energy absorbed due to the impact and is the procedure followed in standard Charpy V- notch testing. Hence, the correct answer is Option A, **Toughness test**.

53. Problem

Correct Answer: (C) W18X106

Explanation:

Per IBC 2018 page 363, Table 1604.3, For floor beams subjected to live loads, the deflection limit is $L/360$

$$\Delta = 25 \text{ ft} \times 12 \text{ (in/ft)} / 360$$

$$\Delta = 0.833 \text{ in}$$

From AISC steel manual 15th Ed, page 3-210, Table 3-23 case 7,

$$\Delta = P \cdot L^3 / (48 \cdot E \cdot I)$$

$$I = P \cdot L^3 / (48 \cdot E \cdot \Delta)$$

$$I = 80 \text{ kip} \cdot (25 \cdot 12 \text{ in})^3 / (48 \cdot 29000 \text{ ksi} \cdot 0.833 \text{ in})$$

$$I = 1862.8 \text{ in}^4$$

From Steel manual page 3-29, Table 3-3

The lightest section W18 that satisfies the requirement is

W18X106 with $I_x = 1910 \text{ in}^4$.

Hence Option C is the answer

W24×84	2370
W18×119	2190
W14×176	2140
W12×210	2140
W24×76	2100
W21×93	2070
W18×106	1910
W14×159	1900
W12×190	1890
W24×68	1830
W21×83	1830
W18×97	1750

54. Problem

Correct Answer: (C) 14.9 kip*ft

Explanation:

Ref to AISC steel manual 15th ed., part AISC 360-16, page 16.1-65, section F11

$$F_y = 50 \text{ ksi}$$

$$E = 29000 \text{ ksi}$$

$$L_b = 3 \text{ ft} = 36 \text{ in}$$

$$D = 5 \text{ in}$$

$$t = 1 \text{ in}$$

Section modulus

$$S = b \cdot d^2 / 6 = 1 \text{ in} \cdot 5 \text{ in} \cdot 5 \text{ in} / 6 = 4.167 \text{ in}^3$$

$$Z = b \cdot d^2 / 4 = 1 \text{ in} \cdot 5 \text{ in} \cdot 5 \text{ in} / 4 = 6.25 \text{ in}^3$$

Elastic and plastic yielding moments

$$M_y = F_y \cdot S = 50 \text{ ksi} \cdot 4.167 \text{ in}^3 = 208.3 \text{ kip} \cdot \text{in}$$

$$M_p = F_y \cdot Z = 50 \text{ ksi} \cdot 6.25 \text{ in}^3 = 312.5 \text{ kip} \cdot \text{in}$$

Check lateral torsional buckling

$$Lb*d/t^2 = 36 \text{ in} * 5 \text{ in} / (1 \text{ in})^2 = 180$$

$$0.08E/Fy = 0.08*29000 \text{ ksi}/50 \text{ ksi} = 46.4$$

$$1.9*E/Fy = 1.9*29000 \text{ ksi}/50 \text{ ksi} = 1102$$

$$\text{Thus, } 0.08E/Fy < Lb*d/t^2 < 1.9*E/Fy$$

From page 16.1-66, equation F11-2

$$Mn = Cb * [1.52 - 0.274*(Lb*d/t^2) * (Fy/E)] * My$$

$$Mn = 1 * [1.52 - 0.274*180*50 \text{ ksi}/29000 \text{ ksi}] * 208.3 \text{ kip*in}$$

$$Mn = 298.9 \text{ kip*in} (< Mp \text{ OK})$$

Allowable flexural strength

$$= Mn/\Omega_b$$

$$= 298.9 \text{ kip*in} / 1.67$$

$$= 178.98 \text{ kip*in}$$

$$= \mathbf{14.9 \text{ kip*ft}}$$

Hence the correct answer is option C.

55. Problem

Correct Answer: (D) 36.75 kip*ft

Explanation:

$$\text{Effective depth, } d = 24 \text{ in} - 4 \text{ in} = 20 \text{ in}$$

Ref to TMS 402/602-16 page C-114, section 8.3.3.1

Allowable stress in 60 ksi rebar , $f_s = 32000 \text{ psi}$

From TMS 402/602-16, page C-64 Table CC-6.1.2,

Area of #9 bar, $A_s = 1 \text{ in}^2$

Max tension force based on max rebar stress = $A_s * f_s$

$$T = 1 \text{ in}^2 * 32000 \text{ psi} = 32000 \text{ lbs}$$

Ref to TMS 402/602-16 page C-115, section 8.3.4.2.2

Allowable stress in masonry due to flexural compression = $0.45f'_m$

$$f_b = 0.45*2000 \text{ psi}$$

$$f_b = 900 \text{ psi}$$

Max compression force based on max masonry stress (for triangular distribution of stress)

$$C = 0.5 * f_b * (0.366*d) * \text{width of section}$$

$$C = 0.5*900 \text{ psi} * 0.366* 20 \text{ in} * 7.625 \text{ in}$$

$$C = 25116.75 \text{ lbs}$$

$C_{max} < T_{max}$, so stress in masonry controls the moment strength.

Couple arm $j \cdot d = 0.878 \cdot d = 0.878 \cdot 20 \text{ in} = 17.56 \text{ in}$

Allowable moment = $C \cdot j \cdot d$

= $25116.75 \text{ lbs} \cdot 17.56 \text{ in}$

= $441050.13 \text{ lbs} \cdot \text{in}$

= **36.75 kip·ft**

Correct answer is Option D.

56. Problem

Correct Answer: (B) I and II

Explanation:

From TMS 402/602-16, page C-16, definition of deep beam

For simple span, $l_{eff}/d < 2$

For continuous span $l_{eff}/d < 3$

Option I simple span, $l_{eff}/d = 10 \cdot 12 / 65 = 1.84 < 2$

Option II continuous span, $l_{eff}/d = 15 \cdot 12 / 65 = 2.76 < 3$

For option III, effective length is not provided so its type cannot be determined.

Correct answer is Option B, **I and II**.

57. Problem

Correct Answer: (A) 10.2 in³

Explanation:

From NDS-2018, page 11, Table 2.3.2, for snow load, $C_d = 1.15$

For dry condition and normal temperature, $C_m = 1$ and $C_t = 1$

Since it is loaded along the thickness and not along the flat (width) face,

Flat use factor, $C_{fu} = 1$

From NDS-2018, page 30, section 4.3.9

For rafter spaced less than 24 in, $C_r = 1.15$

From NDS-2018 supplement page 35, Table 4A

For Eastern white pine, select structural, $F_b = 1250 \text{ psi}$

From NDS-2018, page 29, Table 4.3.1

$F'_b = F_b \cdot C_d \cdot C_m \cdot C_t \cdot C_L \cdot C_f \cdot C_{fu} \cdot C_i \cdot C_r$

$$F'b = 1250 \text{ psi} * 1.15 * 1 * 1 * 1 * 1 * 1 * 1 * 1.15 = 1653.125 \text{ psi}$$

$$\text{Moment acting on the beam} = w * L^2 / 8$$

$$= 50 \text{ plf} * 15 \text{ ft} * 15 \text{ ft} / 8 = 1406.25 \text{ lb-ft} = 16875 \text{ lb-in}$$

$$\text{Required section modulus} = M / F'b$$

$$= 16875 \text{ lb-in} / 1653.125 \text{ psi} = \mathbf{10.2 \text{ in}^3}$$

Correct answer is option A.

58. Problem

Correct Answer: (C) 160 kips

Explanation:

From ACI 318-14, page 341 Table 21.2.1,

strength reduction factor for shear, $\Phi = 0.75$

From ACI 318-14, page 362, Section 22.6.4.1,

The critical perimeter for interior columns with drop panels is located at distance $d/2$ from the edge of the drop panel.

$$\text{Effective depth, } d = 6 \text{ in} - 1.25 \text{ in} = 4.75 \text{ in}$$

$$b_o = 4 * (16 \text{ in} + 12 \text{ in} + 12 \text{ in} + d/2 + d/2)$$

$$b_o = 179 \text{ in}$$

Design two way shear strength

$$= \Phi * v_c * b_o * d$$

$$= 0.75 * 250 \text{ psi} * 179 \text{ in} * 4.75 \text{ in}$$

$$= 159421.87 \text{ lbs}$$

$$\approx \mathbf{160 \text{ kips}}$$

Correct answer is Option C.

59. Problem

Correct Answer: (C) 440 kips

Explanation:

From AISC steel manual 15th ed., page 16.1-570, table C-A-7.1,

Effective length factor for fixed-fixed column, $k = 0.65$

Effective length

$$k * L = 0.65 * 19.25 \text{ ft} * (12 \text{ in/ft})$$

$$k * L = 150.15 \text{ in}$$

From steel manual page 1-97, table 1-12,

Area of HSS 8X8X1/2

$$A_g = 13.5 \text{ in}^2$$

Radius of gyration, $r = 3.04$

$$K \cdot l / r = 150.15 \text{ in} / 3.04 \text{ in}$$

$$K \cdot l / r = 49.4$$

From steel manual page 4-230, table 4-14

For $F_y = 70 \text{ ksi}$ and $L/r = 49$

$$F_{cr} / \Omega = 32.8 \text{ ksi}$$

Hence, allowable axial load

$$= (F_{cr} / \Omega) \cdot A_g$$

$$= 32.8 \text{ ksi} \cdot 13.5 \text{ in}^2$$

$$= 442.8 \text{ kips}$$

Closest answer is Option C, **440 kips**.

60. Problem

Correct Answer: (B) **1.125 in**

Explanation:

From AISC steel manual page 15th ed. Page 14-5, eq 14-2, eq 14-3 and eq 14-4

$$m = (N - 0.95 \cdot d) / 2 = (15 \text{ in} - 0.95 \cdot 11 \text{ in}) / 2 = 2.275 \text{ in}$$

$$n = (B - 0.8 \cdot b_f) / 2 = (15 \text{ in} - 0.8 \cdot 12 \text{ in}) / 2 = 2.7 \text{ in}$$

$$n' = [\sqrt{(d \cdot b_f)}] / 4 = [\sqrt{(11 \cdot 12)}] / 4 = 2.872 \text{ in}$$

Since concrete strength is not given, take $\lambda = 1$

Bearing length $l = \text{maximum between } m, n, \text{ and } \lambda \cdot n'$

$$l = 2.872 \text{ in}$$

From Page 14-6, equation 14-7a

$$t_{\min} = l \cdot \sqrt{2 \cdot P_u / (0.9 \cdot F_y \cdot B \cdot N)}$$

$$t_{\min} = 2.872 \text{ in} \cdot \sqrt{2 \cdot 650 \text{ kips} / (0.9 \cdot 50 \text{ ksi} \cdot 15 \text{ in} \cdot 15 \text{ in})}$$

$$t_{\min} = 1.03 \text{ in}$$

Hence 1 in plate doesn't work, provide **1.125 in** plate.

Correct answer is option B.

61. Problem

Correct Answer: (C) 1.3

Explanation:

Length of column, $L_c = 12$ ft

Length of beam, $L_b = 15$ ft

From AISC Steel Manual 15th ed. page 1-28 and page 1-29, table 1-1

For W10X60, $I_b = 341$ in⁴

From page 1-26 and 1-27, table 1-1

For W12X40, $I_c = 307$ in⁴

For rigid frames, from page 16.1-571 to 16.1-573

For base of column, $G_b = 1$ for rigidly connected to base from page 16.1-573

For top of column, from page 16.1-572

$$G_a = \frac{\sum(E_c I_c / L_c)}{\sum(E_b I_b / L_b)}$$

Since E is equal, it cancels out

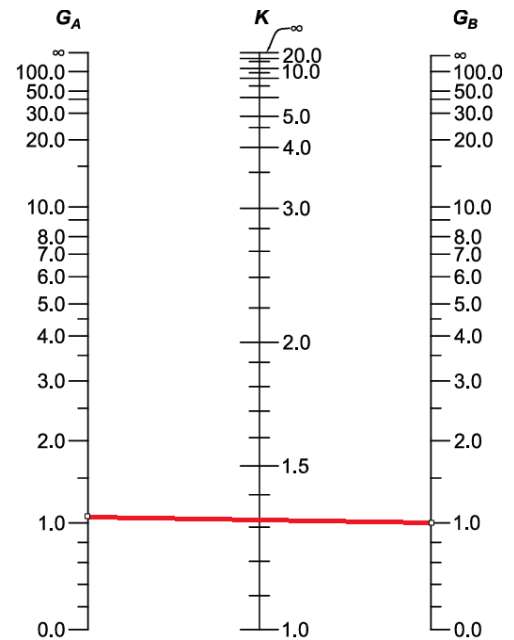
$$G_a = (307/12) / (341/15)$$

$$G_a = 1.125$$

From Fig C-A-7.2 for side sway uninhibited,

$$K = 1.32$$

Hence the correct answer is Option C, **1.3**.



62. Problem

Correct Answer: (A) 22 kips /ft

Explanation:

Per TMS 402/602-16 page C-57, section 5.3.1.4 (b)

Spacing of lateral ties should not exceed the least dimension of the member.

Here #3 ties are spaced at 12 in c/c which is greater than 7.625 in

Thus rebars are not effectively braced per requirements

Effective area of rebars, $A_{st} = 0$

Calculate available axial length per foot length

From page C-115, section 8.3.4.2.1 (b), eq 8-19

$$P_a = (0.25 \cdot f'_m \cdot A_n + 0.65 \cdot A_{st} \cdot F_s) \cdot (70r/H)^2$$

$$P_a = 0.25 \cdot 2000 \text{ psi} \cdot [12 \text{ (in/ft)} \cdot 7.625 \text{ in}] \cdot (70/100)^2$$

$$P_a = 22417.5 \text{ lbs /ft}$$

$$P_a \approx \mathbf{22 \text{ kips /ft}}$$

Correct answer is Option A.

63. Problem

Correct Answer: (C) 8

Explanation:

From ACI 318-14, page 156, section 10.6.1.1, maximum area of longitudinal bars permitted

is $0.08 \cdot \text{Gross area}$

$$= 0.08 \cdot 10 \text{ in} \cdot 10 \text{ in}$$

$$= 8 \text{ in}^2$$

From ACI 318-14, page 507, Appendix A

$$\text{Area of \#9 bar} = 1 \text{ in}^2$$

Max number of bars permitted

$$= 8 \text{ in}^2 / 1 \text{ in}^2$$

$$= \mathbf{8}$$

Correct answer is Option C.

64. Problem

Correct Answer: (B) 50 kips

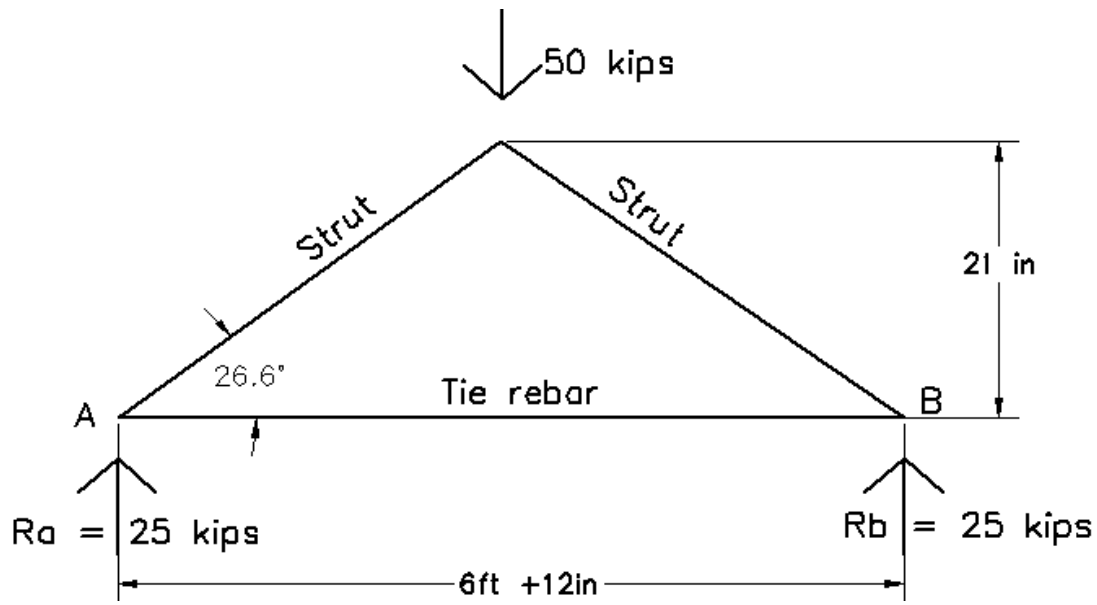
Explanation:

$$d = 21 \text{ in}$$

$$\text{Span of beam} = 6 \text{ ft} + 12 \text{ in}/2 + 12 \text{ in}/2 = 84 \text{ in}$$

$$\Theta = \tan^{-1} [21 \text{ in} / (0.5 \cdot 84 \text{ in})] = 26.6^\circ$$

The given problem can be idealized as a truss with two struts and a tie as follows:



Reaction at each end = 0.5×50 kips = 25 kips due to symmetry

Force in strut, C

Taking sum of vertical forces = 0 at end A

$$R_a - C \sin (26.6^\circ) = 0$$

$$C = R_a / \sin (26.6^\circ)$$

Taking sum of horizontal forces = 0 at end A, Force in horizontal rebar tie 'T'

$$C \cos (26.6^\circ) - T = 0$$

$$T = C \cos (26.6^\circ) = [R_a / \sin (26.6^\circ)] \cos (26.6^\circ) = R_a / \tan (26.6^\circ)$$

$$T = 25 \text{ kips} / 0.5 = \mathbf{50 \text{ kips}}$$

Correct answer is Option B.

65. Problem

Correct Answer: (A) 372 kips

Explanation:

For W12X72, from AISC Steel manual 15th Ed, page 1-26 Table 1-1,

Area, $A_s = 21.1 \text{ in}^2$

From AISC Steel manual 15th edition, page 16.1-91 section 1b

Strength reduction factor for axial compression of composites, $\Omega = 2$

$F_y = 50 \text{ ksi}$

$f'_c = 4 \text{ ksi}$

Area of concrete, $A_c = 16 \text{ in} \times 16 \text{ in} - A_s = 16 \text{ in} \times 16 \text{ in} - 21.1 \text{ in}^2$

$A_c = 234.9 \text{ in}^2$

From page 16.1-91 equation I2-4

$P_{no} = F_y A_s + F_{ysr} A_{sr} + 0.85 f'_c A_c$ (Area of rebar, $A_{sr} = 0$)

$P_{no} = 50 \text{ ksi} \times 21.1 \text{ in}^2 + 0.85 \times 4 \text{ ksi} \times 234.9 \text{ in}^2$

$P_{no} = 1853.7 \text{ kips}$

Given, elastic critical buckling strength, $P_e = 850 \text{ kips}$

$P_{no}/P_e = 1853.7 \text{ kips} / 850 \text{ kips} = 2.181$

From page 16.1-91 equation I2-2

When $P_{no} / P_e \leq 2.25$

$$P_n = P_{no} \left[0.658^{\frac{P_{no}}{P_e}} \right]$$

$$P_n = 1853.7 \text{ kips} \left[0.658^{2.181} \right]$$

$P_n = 744.03 \text{ kips}$

Allowable strength

$P_n / \Omega = 744 \text{ kips} / 2 = \mathbf{372 \text{ kips}}$

Correct answer is Option A.

66. Problem

Correct Answer: (C) 1970 in⁴

Explanation:

The point of max moment to zero moment in a simply supported beam with Uniformly distributed load is half the span length, number of studs, $n = 40/2 = 20$

$$\Sigma Q_n = 16.8 \text{ kip} * 20 = 336 \text{ kips}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-7,

$$a = \Sigma Q_n / (0.85 * f'_c * b) = 336 \text{ kips} / [0.85 * 5.5 \text{ ksi} * 6 \text{ ft} * (12 \text{ in/ft})] = 1 \text{ in}$$

From AISC Steel manual 15th edition, page 3-13, eqn. 3-6,

$y_2 = Y_{con} - a/2$, where Y_{con} is the distance between top of steel beam to top of concrete

$$y_2 = (4 \text{ in} + 2.5 \text{ in}) - 1 \text{ in}/2 = 6 \text{ in}$$

From AISC Steel manual 15th edition, page 3-197, Table 3-20,

For W18X55, $y_2 = 6 \text{ in}$ and $\Sigma Q_n = 336 \text{ kips}$, $I_{LB} = \mathbf{1970 \text{ in}^4}$

Correct answer is Option C.

67. Problem

Correct Answer: (A)

Explanation:

The nominal joint shear strength is not limited to $20\lambda\sqrt{f'_c}$ times joint area and varies per framing conditions (ACI 318-14, page 287, Table 18.8.4.1).

There are no clauses that confirm Option C and D and both are incorrect.

From ACI 318-14, page 286, section 18.8.3.1, joint transverse reinforcement should satisfy the requirements set for columns in the preceding sections except for the case in section 18.8.3.2 which does not apply here. Hence correct answer is Option A.

68. Problem

Correct Answer: (B)

Explanation:

Fully restrained connections have negligible relative rotation between the members and are simpler to analyze, compared to partially restrained connections which add additional relative rotational indeterminacy to the analysis. Option A is true.

Option C and D are true.

Option B is not true. Partially restrained connections, although transmit moments partially, do not allow unrestrained relative rotation. They have partial fixity.

Hence, Option B is the correct answer.

69. Problem

Correct Answer: (C) **105 kips**

Explanation:

Vertical Spacing of bolts, $s = 3$ in

Number of bolts in a vertical row, $n = 6$

Eccentricity, $e = 7.25$ in + $(5.5$ in $/2) = 10$ in

From AISC steel manual 15th ed. Page 7-43, Table 7-8,

$C = 5.23$

Thus allowable load in the connection

$P = C * \text{allowable bolt shear strength}$

$P = 5.23 * 20$ kips = 104.6 kips $\approx$ **105 kips**

Correct answer is Option C.

70. Problem

Correct Answer: (D) 55 ksi

Explanation:

Total number of bolts, $n = 9$

Shear Load per bolt, $P = 250 \text{ kips}/n$

$= 250 \text{ kips}/9 = 27.78 \text{ kips}$

From AISC steel manual 15th ed. , page 16.1-130, Table J3.3, nominal hole size for standard holes with $3/4 \text{ in.}$ diameter bolts $= 13/16 \text{ in.}$

Thickness of plate, $t = 5/8 \text{ in}$

Projected area of bolt hole for bearing, $A = 13/16 \text{ in} * 5/8 \text{ in}$

$A = 0.508 \text{ in}^2$

Bearing stress $= P/A$

$= 27.78 \text{ kips} / 0.508 \text{ in}^2 \approx \mathbf{55 \text{ ksi}}$

Correct answer is Option D.

71. Problem

Correct Answer: (A) 21.1 kips

Explanation:

From AISC steel manual, page 10-91, the single plate shear connection meets all requirements for Table 10-10a to be used.

From Page 10-94, Table 10-10 a, for $3/8 \text{ inch}$ plate, $3/4 \text{ inch}$ bolt and $n = 2$,

For Group A bolts (A325) with threads not in shear plane,

Allowable strength $= \mathbf{21.1 \text{ kips}}$

Correct answer is Option A.

72. Problem

Correct Answer: (C) 145.8 kips

Explanation:

Diameter of pin, $d = 3 \text{ in}$

Thickness of plate, $t = 1.5 \text{ in}$

From AISC steel manual 15th ed, page 16.1-140 , Equation J7-1

$R_n = 1.8 * F_y * \text{projected area in bearing}$

$R_n = 1.8 * 36 \text{ ksi} * 3 \text{ in} * 1.5 \text{ in}$

$$R_n = 291.6 \text{ kips}$$

$$\text{Allowable bearing strength} = R_n / \Omega$$

$$= 291.6 \text{ kips} / 2 = \mathbf{145.8 \text{ kips}}$$
 Correct answer is Option C.

73. Problem

Correct Answer: (B) 3.25 in

Explanation:

$$\text{Total load} = \text{Dead} + \text{Live} = 10 \text{ kips} + 25 \text{ kips} = 35 \text{ kips}$$

$$\text{Size of } 3/8 \text{ inch weld in } 16\text{th of an inch}, 3/8 \text{ in} = 6/16 \text{ in}. \text{ Hence } D = 6$$

$$\text{From AISC steel manual 15th ed. Page 8-8, eqn. 8-2b, } R_n/\Omega = 0.928 * D * l$$

$$35 \text{ kips} = 0.928 * 6 * l$$

$$l = 6.28 \text{ in}$$

$$L_w = l/2 = 6.28 \text{ in} / 2 = 3.14 \text{ in}$$

Hence the correct answer is option B, **3.25 in**.

74. Problem

Correct Answer: (D) 48 in.

Explanation:

From ACI 318-14 page 507, Appendix A, diameter of #8 bar, $d_b = 1 \text{ in}$

From ACI 318-14, page 417, Table 25.4.2.4

$\Psi_t = 1.0$ for bottom bars with less than 12 in. of concrete below.

From ACI 318-14, page 415, Table 25.4.2.2, development length for #8 bar

$$L_d = (f_y * \Psi_t * \Psi_e) / (20 * \lambda * \sqrt{f'_c}) * d_b$$

$$L_d = (60000 * 1 * 1) / (20 * 1 * \sqrt{4000}) * 1 \text{ in} = 47.4 \text{ in}$$

From ACI 318-14 page 142, section 9.7.3.3, from point of inflection, bars should extend at least by a length that is greater of 'd' or $12 * d_b$

$$d = 21 \text{ in} \quad 12 * d_b = 12 \text{ in}$$

Thus, point of inflection i.e 2.5ft (30 in) is sufficiently away from the face of support.

From ACI 318-14 page 143, section 9.7.3.8.2, at least one-fourth of the maximum positive moment reinforcement shall extend along the beam bottom into the support. If the beam is part of the primary lateral-load-resisting system, bars shall be anchored to develop 'fy' at the face of the support.

Hence required extension = l_d . Correct answer is Option D, **48 in**.

75. Problem

Correct Answer: (C) 300 plf

Explanation:

From NDS 2018 page 11, Table 2.3.2, for dead loads, $C_d = 0.9$

From NDS 2018 supplement page 14, Table 1B

Dressed size of 4X8 = 3.5 in X 7.25 in

From NDS 2018, page 99, Table 12E

For Douglas Fir-Larch with side member 3.5 in thick, and 1 inch diameter bolt, loaded perpendicular to grain, $Z_{\perp} = 1330$ lbs

Since all adjustment factors are 1 except load factor,

$$Z_{\perp}' = C_d * Z_{\perp} = 0.9 * 1330 \text{ lbs} = 1197 \text{ lbs}$$

Allowable dead load = $Z_{\perp}' / \text{spacing}$

$$= 1197 \text{ lbs} / 4 \text{ ft} \approx \mathbf{300 \text{ plf}}$$

Correct answer is Option C.

76. Problem

Correct Answer: (D) 14.24 ksf

Explanation:

From NCEES PE handbook page 91, Bearing capacity factors,

For $\Phi = 25^\circ$, $N_c = 20.7$ $N_q = 10.7$ $N_{\gamma} = 10.9$

From NCEES PE handbook page 90, section 3.4.2

$$q = \gamma * D_f = 115 \text{ pcf} * 5 \text{ ft} = 575 \text{ psf} \quad (\text{Assuming surcharge } Q_{\text{appl}} = 0)$$

$$q_{\text{ult}} = c * N_c + q * N_q + 0.5 * \gamma * B_f * N_{\gamma}$$

$$q_{\text{ult}} = 300 \text{ psf} * 20.7 + 575 \text{ psf} * 10.7 + 0.5 * 115 \text{ pcf} * 3 \text{ ft} * 10.9$$

$$q_{\text{ult}} = 14242.75 \text{ psf} = \mathbf{14.24 \text{ ksf}}$$

Correct answer is Option D.

77. Problem

Correct Answer: (C) 2.9 ksf

Explanation:

The objective of a strap footing is to use a rigid strap to connect two footings so as to achieve equal soil pressure under both footings. The strap beam does not provide any bearing resistance.

Calculate weight of strap and footings.

$$\text{Self Weight, } W = (4\text{ft} \times 6\text{ft} \times 2.5\text{ft} + 5\text{ft} \times 10\text{ft} \times 2.5\text{ft} + 2\text{ft} \times 2\text{ft} \times 10\text{ft}) \times 0.15 \text{ kcf}$$

$$W = 33.75 \text{ kips}$$

$$P_a = 58 \text{ kips}$$

$$P_b = 121 \text{ kips}$$

$$\text{Total vertical load, } P = W + P_a + P_b$$

$$= 33.75 \text{ kips} + 58 \text{ kips} + 121 \text{ kips}$$

$$= 212.75 \text{ kips}$$

$$\text{Total bearing area, } A = \text{area of footing A} + \text{area of footing B}$$

$$= 4 \text{ ft} \times 6\text{ft} + 5 \text{ ft} \times 10 \text{ ft}$$

$$= 74 \text{ ft}^2$$

$$\text{Bearing pressure under footings} = P/A$$

$$= 212.75 \text{ kips} / 74 \text{ ft}^2$$

$$= \mathbf{2.9 \text{ ksf}}$$

Correct answer is Option C.

78. Problem

Correct Answer: (C) 21 kip*ft

Explanation:

$$\text{Bearing pressure, } q = 50 \text{ kips} / (6\text{ft} \times 6\text{ft}) = 1.39 \text{ ksf}$$

From ACI 318-14 page 191, Table 13.2.7.1, Location of critical section for Mu is the face of the column

Lever arm is the distance between edge of the footing to face of the column

$$L = 6\text{ft} / 2 - 18\text{in} / 2$$

$$L = 2.25 \text{ ft}$$

$$\text{Moment acting at the face of the column due to bearing pressure, } q$$

$$= q \times L \times L / 2$$

$$= 1.39 \text{ ksf} \times 2.25 \text{ ft} \times 2.25 \text{ ft} / 2$$

$$= 3.5 \text{ kip*ft/ft}$$

For total width of 6ft,

$$\text{Total flexural strength required} = 3.5 \text{ kip*ft/ft} * 6\text{ft} = \mathbf{21 \text{ kip*ft}}$$

The correct answer is Option C.

79. Problem

Correct Answer: (B)

Explanation:

Group effects and efficiency is required depending on the center to center spacing of piles.

Option A, C and D all represent generalized scenarios with may or may not warrant the analysis for group effects.

From IBC 2018 page 445, section 1810.2.5, analysis shall include group effects if the spacing of piles is less than three times least horizontal dimension the piles. Hence , it is required if the diameter is more than 1/3rd of the spacing. Option B is the correct answer.

80. Problem

Correct Answer: (D) **22152 lb*ft/ft**

Explanation:

The critical section of heel for flexure is at the face of the stem as marked in the figure with red.

The bearing pressure under the heel is negligible.

So the loads acting on the heel are self weight of wall, surcharge and weight of soil above the stem.

From ACI 318-14, page 60, section 5.3.8, load factor for earth pressure = 1.6

From ACI 318-14, page 58, Table 5.3.1, load factor for dead load = 1.2

$$\text{Self weight of wall} = 150 \text{ pcf} * 15 \text{ in} = 150 \text{ pcf} * 1.25 \text{ ft} = 187.5 \text{ psf}$$

$$\text{Weight of soil above the heel} = 120 \text{ pcf} * (11.5 \text{ ft} - 15 \text{ in}) = 120 \text{ pcf} * 10.25\text{ft} = 1230 \text{ psf}$$

$$\text{Surcharge} = 360 \text{ psf}$$

$$\text{Moment arm for heel} = 4\text{ft} ,$$

Sum of moments from edge of the heel to critical section with appropriate load factors,

$$= 1.2 * 187.5 \text{ psf} * 4\text{ft} * 4\text{ft}/2 + 1.6 * (360 \text{ psf} + 1230 \text{ psf}) * 4\text{ft} * 4\text{ft}/2$$

$$= \mathbf{22152 \text{ lb*ft/ft}}$$

Correct answer is option D.

